



# Advanced Artificial Intelligence

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# Outline covered

- Decision Making
  - Utility Theory and Functions
  - Decision Networks
  - Sequential Decision Problems
- Reinforcement Learning
  - Basic Ideas and Applications
  - Formalization, Markov Decision Processes
  - Different forms of RL: Passive and Active
  - Learning optimal strategies\*
- Probabilistic Reasoning over Time
  - POMDPs, Hidden Markov Models
  - Time-series Analysis: Basic Concepts
  - Kalman Filters
    - Simple 1D and multidimensional filtering

## Evaluation

- Quiz – 1 (15%)
- Quiz – 2 (15%)
- Quiz – 3 (15%)
- Coding Assignment (15%)
  - Programming Languages?
- Major Exam (40%)

## Reference Texts

- *Artificial Intelligence: A Modern Approach* - Stuart Russel, and Peter Norvig
- *Sutton and Barto, Intro to RL 2014, 2015*
- *Relevant papers and videos*

# Probability Recap

Random Variables and probability:

$$X, P(X)$$

Conditional Probability:

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

Product Rule:

$$P(X, Y) = P(X|Y)P(Y)$$

Independence between 2 R.Vs X and Y:

$$P(X, Y) = P(X)P(Y)$$

Baye's Inference:

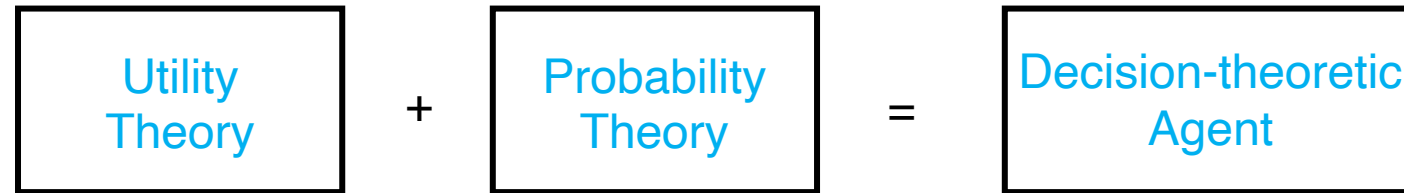
$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Expected value of a R.V. X:

$$E(X) = \sum_{x \in X} x \cdot P(X = x)$$

# Utility Theory and Functions

Concept of making decisions

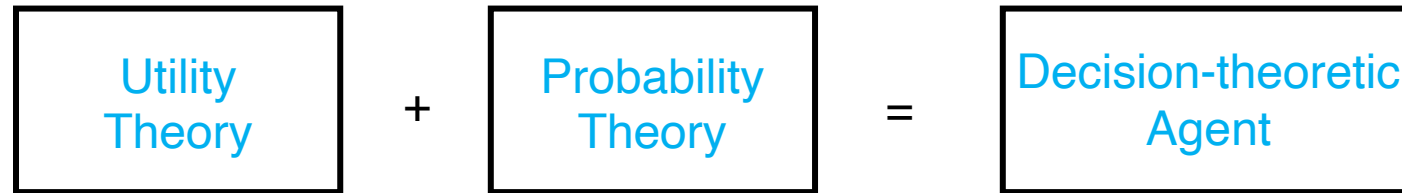


**Decision theory:** Choosing actions based on desirability of immediate outcomes

$$P(\text{RESULT}(a) = s' \mid a, e)$$

# Utility Theory and Functions

Concept of making decisions

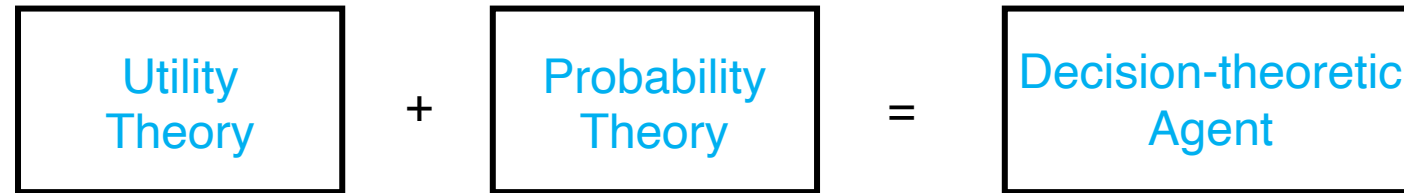


**Agent's Preferences:** Captured by a Utility function (average by expected utility)

$$EU(a|\mathbf{e}) = \sum_{s'} P(\text{RESULT}(a) = s' | a, \mathbf{e}) U(s')$$

# Utility Theory and Functions

Concept of making decisions



**Agent's Preferences:** Maximum Expected Utility (MEU)

$$action = \underset{a}{\operatorname{argmax}} EU(a|\mathbf{e})$$

# Utility Theory and Functions

Some examples to illustrate Utility-based view

- **Medical Decision-Making**

- Scenario: A patient must choose between two treatments:
- Treatment A: 90% chance of full recovery, 10% chance of death
- Treatment B: 100% chance of living, but with a chronic condition

- **Utility-based View**

- The patient assigns a **higher utility to full recovery**, but a very **low utility to death**.
- If they **highly fear death**, even a **small chance may make B more preferable**, despite the guaranteed chronic condition.

# Utility Theory and Functions

Some examples to illustrate Utility-based view

- **Car Insurance**

- Scenario: A person has a bike worth ₹100,000. There's a 1% chance it will be totaled this year.
- Decision:
  - Pay ₹2000 now for insurance
  - Risk losing ₹100,000 if an accident happens

- **Utility-based View**

- Expected monetary loss:  $0.01 \times ₹100,000 = ₹1000$
- Yet they pay ₹2000 for insurance.
- **Utility loss from a ₹100,000 hit is much larger** than the utility loss from a **₹2000 expense**. The person is risk-averse, valuing peace of mind more than raw expected value.



# Utility Theory and Functions

Some examples to illustrate Utility-based view

- **Career Choice**

- Scenario: A student must choose between:
- Job A: Pays ₹100,000, low stress
- Job B: Pays ₹200,000, but high stress, poor work-life balance

- **Utility-based View**

- The student may assign **greater utility** to free time, health, or enjoyment than extra money.
- Thus, even though B pays more, they might choose A because it offers **higher total utility**.

# Utility Theory and Functions

Some examples to illustrate Utility-based view

- **Restaurant Choice**

- Scenario: You're deciding between:
  - A familiar place with decent food (safe bet)
  - A new exotic place with either excellent food or terrible food (risky bet)

- **Utility-based View**

- Some people might prefer the safe choice due to **higher expected utility of a predictable meal**.
- Others (**risk-seeking**) might go for the new place, hoping for the high reward.

# Utility Theory and Functions

## The essence of MEU

- If an agent acts so as to maximize a utility function that correctly reflects the performance measure, then the agent will achieve the highest possible performance score (averaged over all the possible environments).

## Self-driving Car



- **Performance measure**: Minimize accidents, fuel consumption, and time taken.
- Build a **utility function** that reflects these priorities correctly.
- The car uses **MEU**: computes expected utility of each action
- Chooses the action with the **highest expected utility**.

# Utility Theory and Functions

## Basics of Utility Theory

- To understand rational preferences and how they lead to MEU, let's look at 6 foundational principles.

### Orderability

- For any two outcomes (or lotteries) A and B, a rational agent must either:
- Prefer A over B ( $A \succ B$ ),
- Prefer B over A ( $B \succ A$ ), or
- Be indifferent between them ( $A \sim B$ ).

# Utility Theory and Functions

## Basics of Utility Theory

- To understand rational preferences and how they lead to MEU, let's look at 6 foundational principles.

### Transitivity

- If an agent prefers A over B and B over C, then they must prefer A over C.
- If  $A \succ B$  and  $B \succ C \Rightarrow A \succ C$

# Utility Theory and Functions

## Basics of Utility Theory

- To understand rational preferences and how they lead to MEU, let's look at 6 foundational principles.

### Continuity

- If B is between A and C in preference, then there exists some probability p such that the agent is indifferent between getting B for sure and getting a lottery between A and C.
- If  $A \succ B \succ C$ , then  $\exists p$  such that:
- $[p, A; 1-p, C] \sim B$

# Utility Theory and Functions

## Basics of Utility Theory

- To understand rational preferences and how they lead to MEU, let's look at 6 foundational principles.

### Substitutability

- If an agent is indifferent between A and B, then in any lottery, one can be substituted for the other without changing the agent's preference.
- If  $A \sim B$ , then:
- $[p, A; 1-p, C] \sim [p, B; 1-p, C]$

# Utility Theory and Functions

## Basics of Utility Theory

- To understand rational preferences and how they lead to MEU, let's look at 6 foundational principles.

### Monotonicity

- If an agent prefers A over B, then they should prefer a lottery that has a higher chance of A.
- If  $A \succ B$ , then:
- $[p, A; 1-p, B] \succ [q, A; 1-q, B]$  if and only if  $p > q$

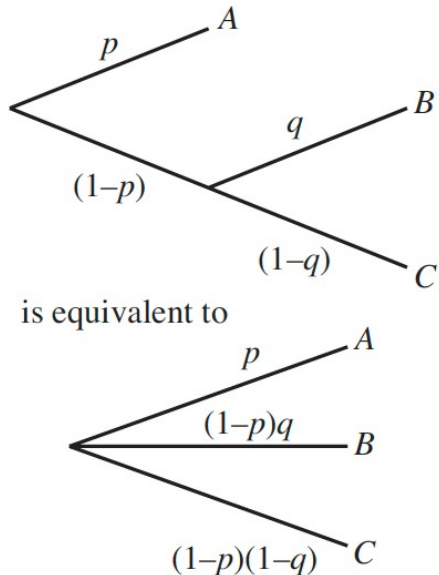


# Utility Theory and Functions

## Basics of Utility Theory

- To understand rational preferences and how they lead to MEU, let's look at 6 foundational principles.

### Decomposability

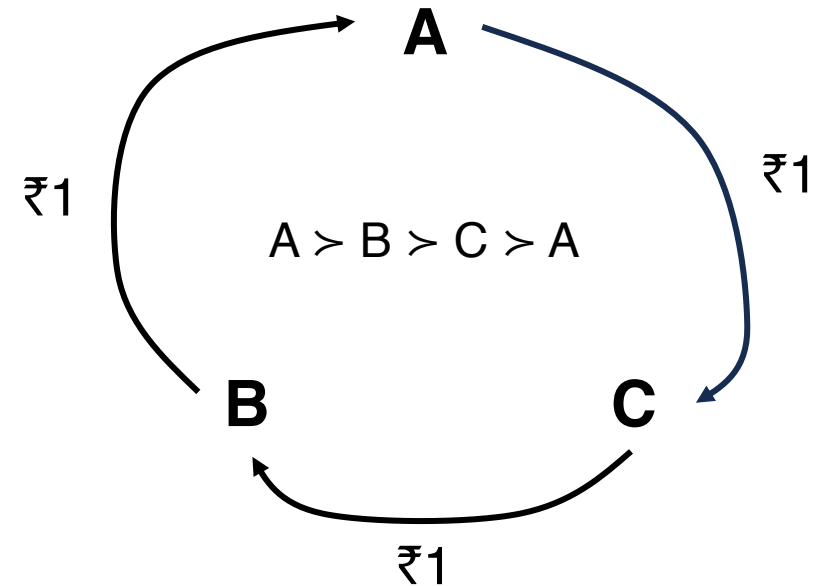


- A compound lottery (a lottery with another lottery as one of the outcomes) can be reduced to an equivalent simple lottery by computing the total probabilities.
- $[p, A; 1-p, [q, B; 1-q, C]] \sim [p, A; (1-p)q, B; (1-p)(1-q), C]$

# Utility Theory and Functions

## Basics of Utility Theory

- To understand rational preferences and how they lead to MEU, let's look at 6 foundational principles.
- **Six axioms** of Utility.
- **Violations** of these axioms means the agent exhibit irrational behaviors in some situations



# Utility Theory and Functions

## Preferences lead to Utility

- **Existence of Utility Function:** There exists a function  $U$ , s.t.

$$\begin{aligned}U(A) > U(B) &\Leftrightarrow A \succ B \\U(A) = U(B) &\Leftrightarrow A \sim B\end{aligned}$$

- **Expected Utility:** sum probability of each outcome multiplied by the utility of that outcome

$$U([p_1, S_1; \dots; p_n, S_n]) = \sum_i p_i U(S_i)$$

- Theorem says Utility exists but is not unique.

$$U'(S) = aU(S) + b$$

# Utility Theory and Functions

## Utility of Money

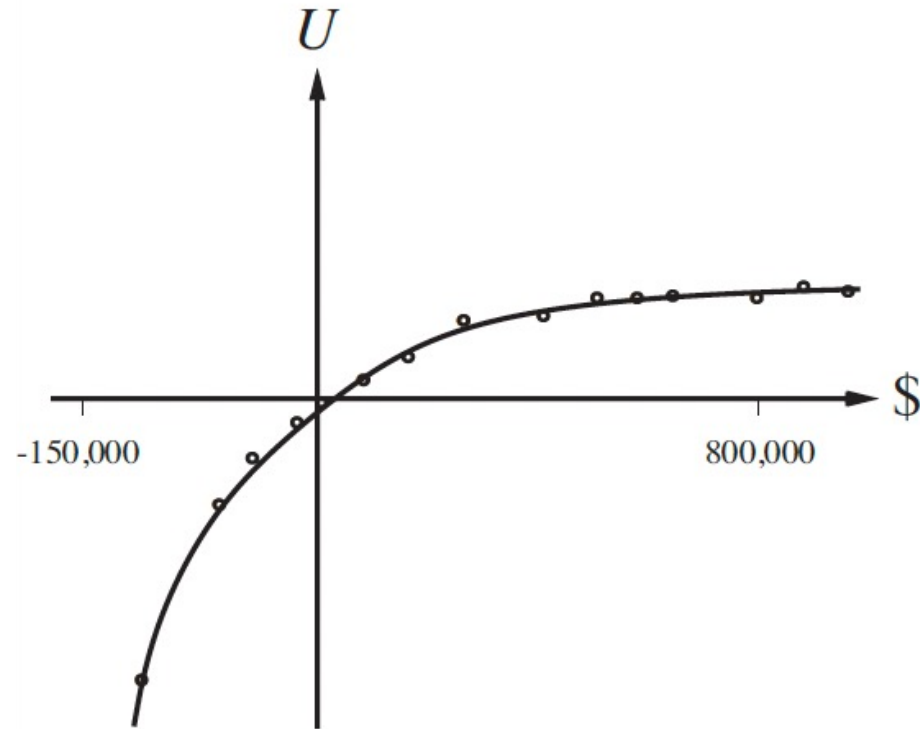
- **Obvious candidate for utility measure:** Money, most people have **monotonic preference** for more money
- Does not mean money behaves like a utility function, take this example:

*“Suppose you have triumphed over the other competitors in a television game show. The host now offers you a choice: either you can take the \$1,000,000 prize or you can gamble it on the flip of a coin. If the coin comes up heads, you end up with nothing, but if it comes up tails, you get \$2,500,000.”*

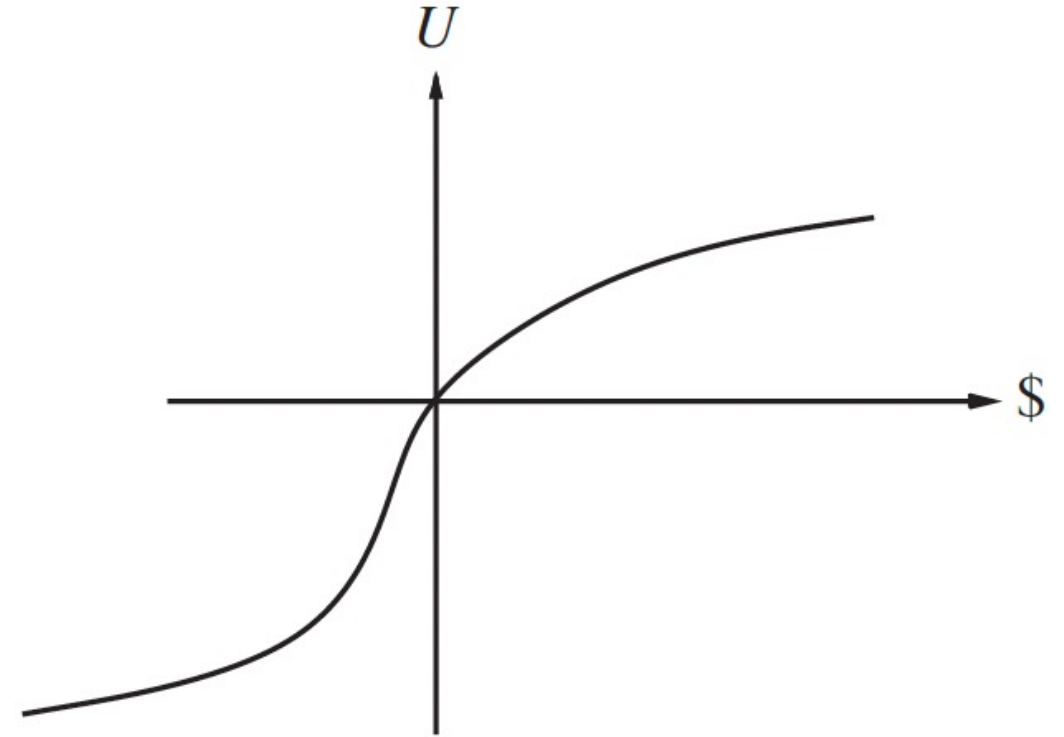
**What will you do?**

# Utility Theory and Functions

## Utility of Money



$$U(S_{k+n}) = -263.31 + 22.09 \log(n + 150,000)$$



# Utility Theory and Functions

## Rational Decisions and post-decision disappointment

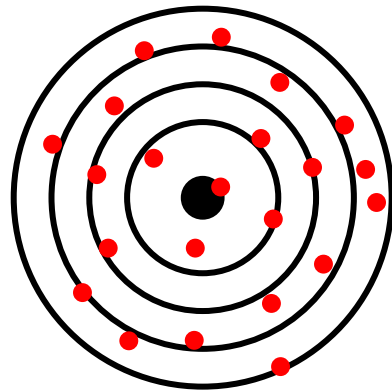
- The rational way to choose the best action:  $a^* = \operatorname{argmax}_a EU(a|e)$
- Maximum Expected Utility (MEU): Choose the action that gives the highest expected utility, given the evidence
- Expected utility (EU) is calculated by combining:
  - The probability of each outcome (from your model)  $\longrightarrow$  In reality, the model oversimplifies
  - The utility (value) of that outcome  $\longrightarrow$  True utilities are intractable, so we use unbiased estimates  $\widehat{EU}$

$$E(\widehat{EU} - EU) = 0$$

# Utility Theory and Functions

## Rational Decisions and post-decision disappointment

- Things still go wrong, how is this possible?
- The real outcome will be significantly worse than we estimated!
- This is where the **optimizers curse** enters:



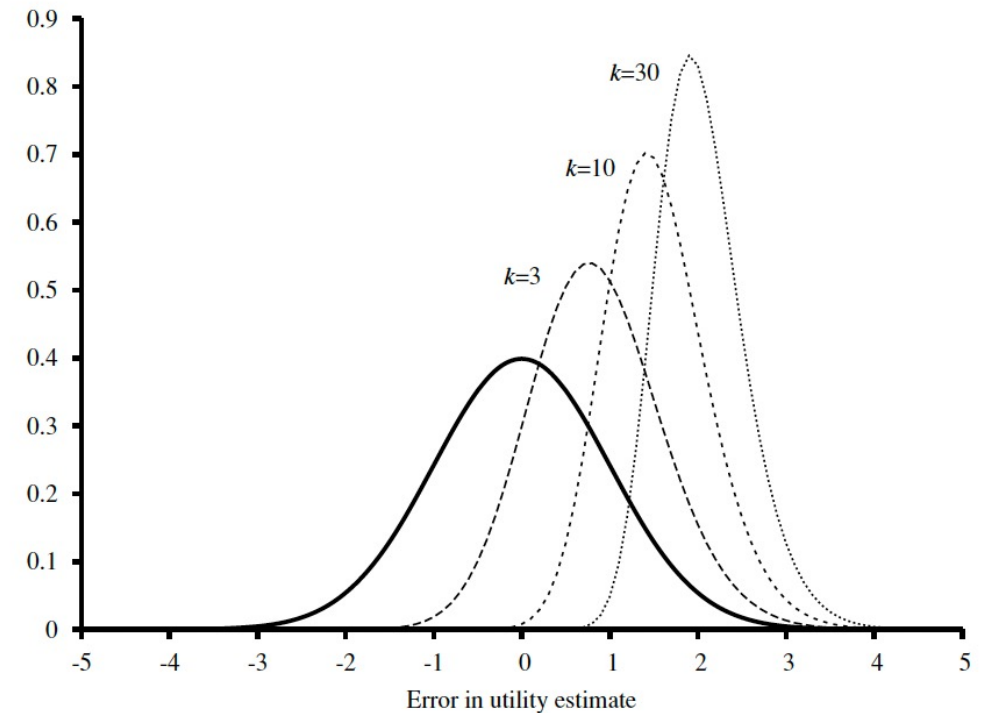
**True Skill** is worse than their **best throw**

- When you pick based on **highest estimated performance**, you often **overestimate the true value** — because you're selecting from among noisy estimates.

# Utility Theory and Functions

## Rational Decisions and post-decision disappointment

- For a more mathematical intuition -
- Let's say you have  $k = 30$  choices
- True Utility of each = 0
- Estimates have a std error of say  $\pm 1$  SD
- Disappointment grows with number of choices ( $k = 3$  vs  $k = 30$ )
- Real world examples – drug testing, investing

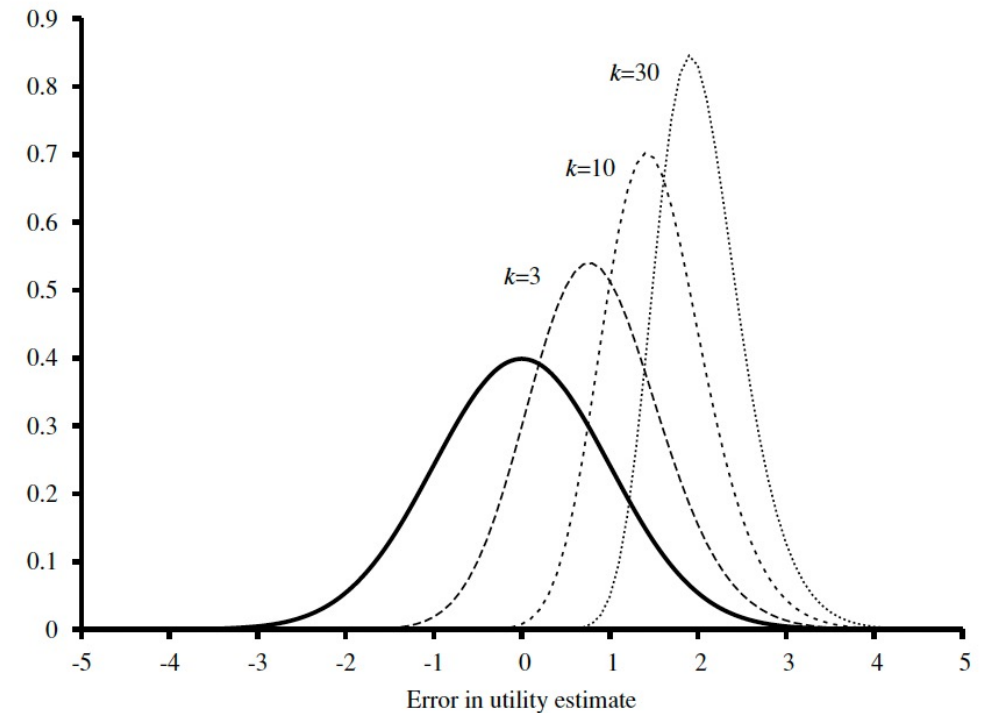




# Utility Theory and Functions

## Rational Decisions and post-decision disappointment

- How do you overcome this?
- Using an explicit probability model
- For instance, the **Baye's rule** helps here
- Start with a **prior estimate of EU**,  $P(EU)$
- Use the **expected utility** as evidence
- Estimate the **posterior distribution**



$$P(EU | \widehat{EU}) \propto P(\widehat{EU} | EU)P(EU)$$

# Utility Theory and Functions

## Human Judgement and Irrationality

- **Decision Theory** – Normative Theory, i.e. talks about how rational agents should act
- **Descriptive Theory** – talks about how actual agent (like humans) really do act
- In essence, the application of these theories would be enhanced if both coincided
- However, evidence is to the contrary – humans are “**predictably irrational**”

# Utility Theory and Functions

## Human Judgement and Irrationality

- For example, let's do the following. What will you prefer?
  - A. 80% chance of \$4000
  - B. 100% chance of \$3000
  - C. 20% chance of \$4000
  - D. 25% chance of \$3000

# Utility Theory and Functions

## Human Judgement and Irrationality

- Now consider this, an urn contains  $\frac{1}{3}$  red balls, and  $\frac{2}{3}$  either black or yellow balls  
What will you prefer?
  - A. \$100 for a red ball
  - B. \$100 for a black ball
  - C. \$100 for a red or yellow ball
  - D. \$100 for a black or yellow ball

# Utility Theory and Functions

## Human Judgement and Irrationality

- For example, let's do the following. What will you prefer?
  - A. 80% chance of \$4000
  - B. 100% chance of \$3000
  - C. 20% chance of \$4000
  - D. 25% chance of \$3000
- Normative analysis disagrees!!
- Assume monotonic preference of money, i.e.  $U(\$0) = 0$
- Allais Paradox

# Utility Theory and Functions

## Human Judgement and Irrationality

- There are many explanations for this effect
- **Certainty Effect** – tend towards options that are certain
- **Distrust** – distrust the legitimacy of the stated probabilities
- **Regret** – account for their emotional state, give up the certain reward for a higher one and then lose it.

# Utility Theory and Functions

## Human Judgement and Irrationality

- Now consider this, an urn contains  $\frac{1}{3}$  red balls, and  $\frac{2}{3}$  either black or yellow balls  
What will you prefer?
  - A. \$100 for a red ball
  - B. \$100 for a black ball
  - C. \$100 for a red or yellow ball
  - D. \$100 for a black or yellow ball
- Ambiguity Aversion
- A gives you a  $\frac{1}{3}$  chance of winning, B could be anywhere between 0 and  $\frac{2}{3}$
- D gives you a  $\frac{2}{3}$  chance, C could be anywhere between  $\frac{1}{3}$  and  $\frac{3}{3}$

# Utility Theory and Functions

## Other interesting Effects

- **Framing Effect:** Exact wording of a decision problem can have big impact on agent's choices
- Treatment has “90% chance of survival” rather than “10% chance of death”
- **Anchoring Effect:** People are more comfortable making relative utility judgements than absolute ones
- Serving a \$200 wine bottle – then serving a \$55 one which seems like a bargain.





# Advanced Artificial Intelligence

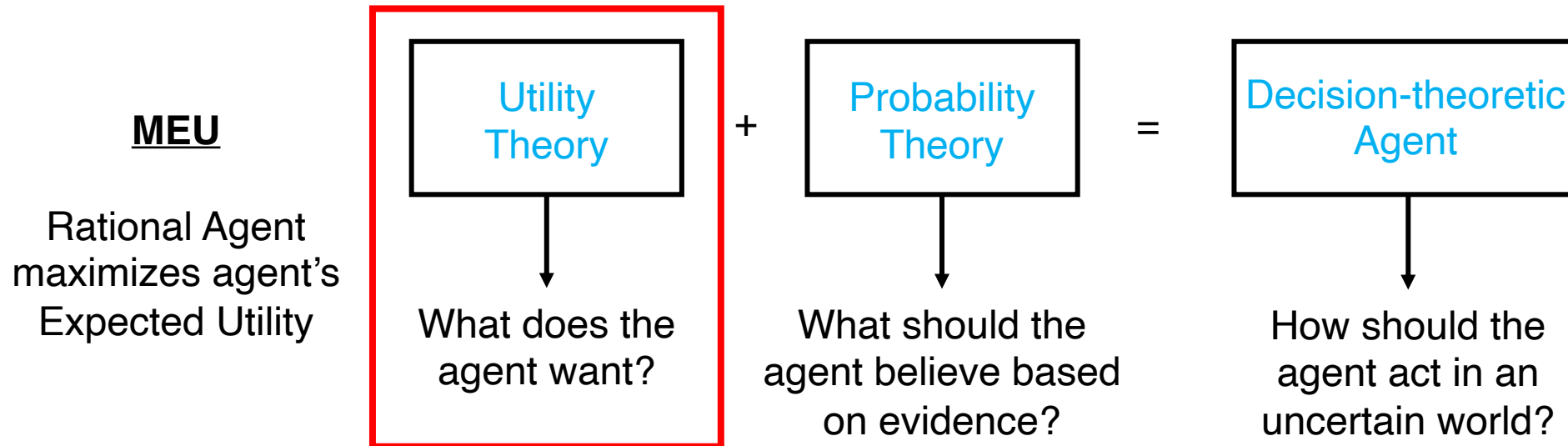
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# Decision Theory and Networks

## Important to reason in an uncertain world

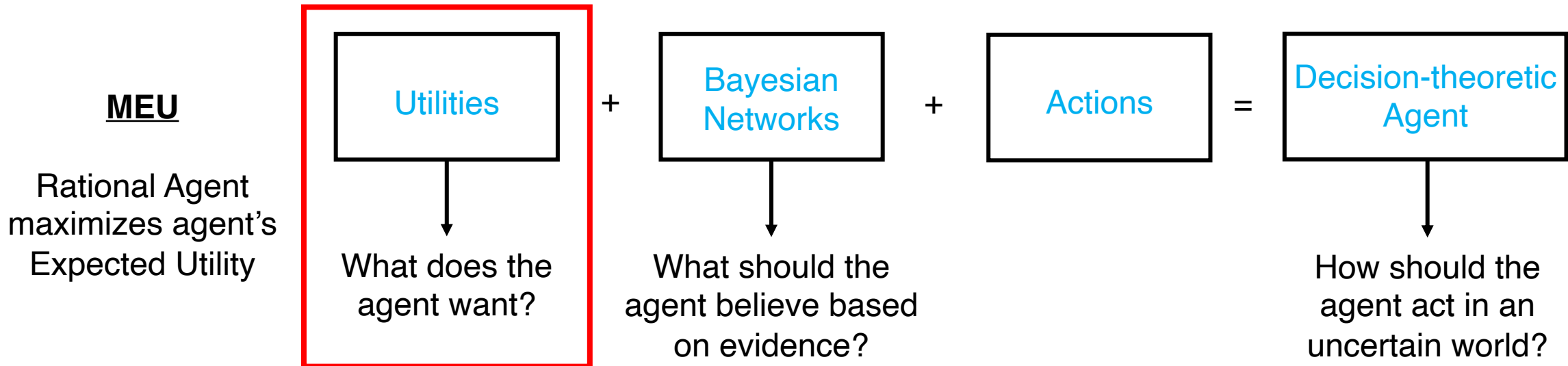
- However, more importantly it is necessary to make decisions in an uncertain world.
- Decision Theory → Acting in an uncertain world



# Decision Theory and Networks

## Important to reason in an uncertain world

- However, more importantly it is necessary to make decisions in an uncertain world.
- Decision Theory → Acting in an uncertain world



# Decision Theory and Networks

## Constructing a Decision Network

- A package pick-up robot – as a running example
- A robot must choose its route to pick up packages. There are two routes: a short one and long one. The short route is a bit treacherous – the robot can encounter an accident. However, the robot can wear protective gear which does not change the probability of an accident but will reduce the damage in case an accident occurs. Unfortunately, the protective gear will add more weight to the robot and slow it down. The aim is to pick up the package as soon as possible while minimizing the damage caused by an accident.
- **What should the robot do?**

# Decision Theory and Networks

## Constructing a Decision Network

- Let's see what are the variables
- What are the random variables here?
  - **A: whether an accident occurs or not**
- What are the decision variables here (actions)?
  - **S: whether the robot chooses the short route**
  - **P: whether the robot uses the protective gear**



Chance Nodes

A diagram showing an oval shape representing a chance node.

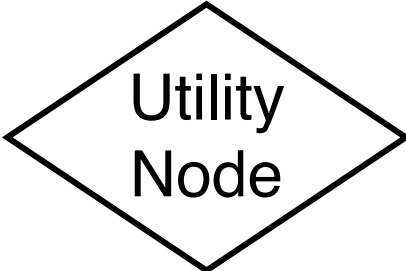
**Random Variables**



Decision Nodes

A diagram showing a rectangular box representing a decision node.

**Decision/Action Variables**



Utility  
Node

A diagram showing a diamond shape representing a utility node.

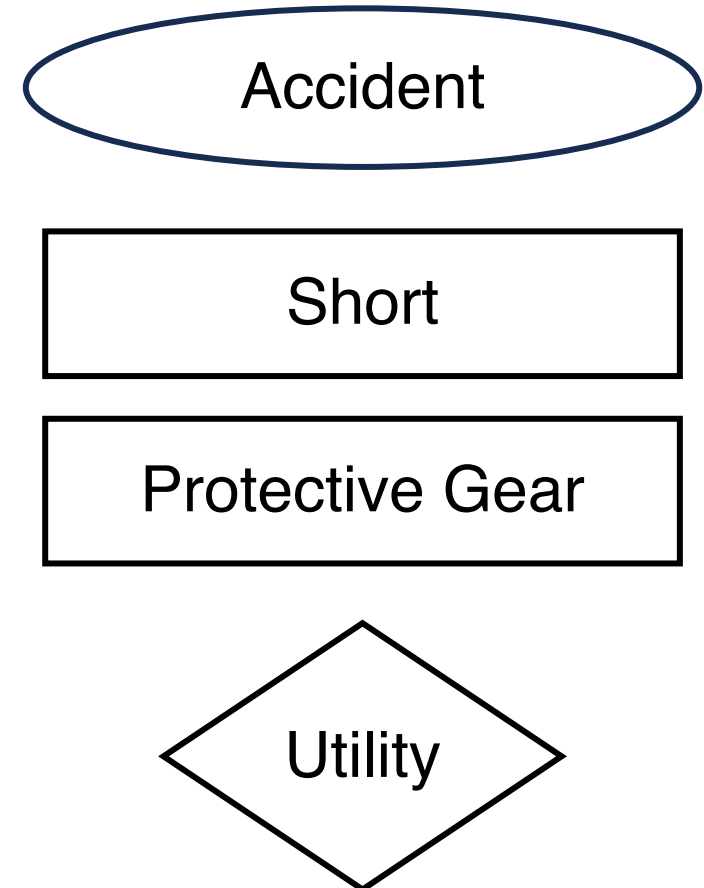
**Agent's Utility Function**

# Decision Theory and Networks

## Constructing a Decision Network

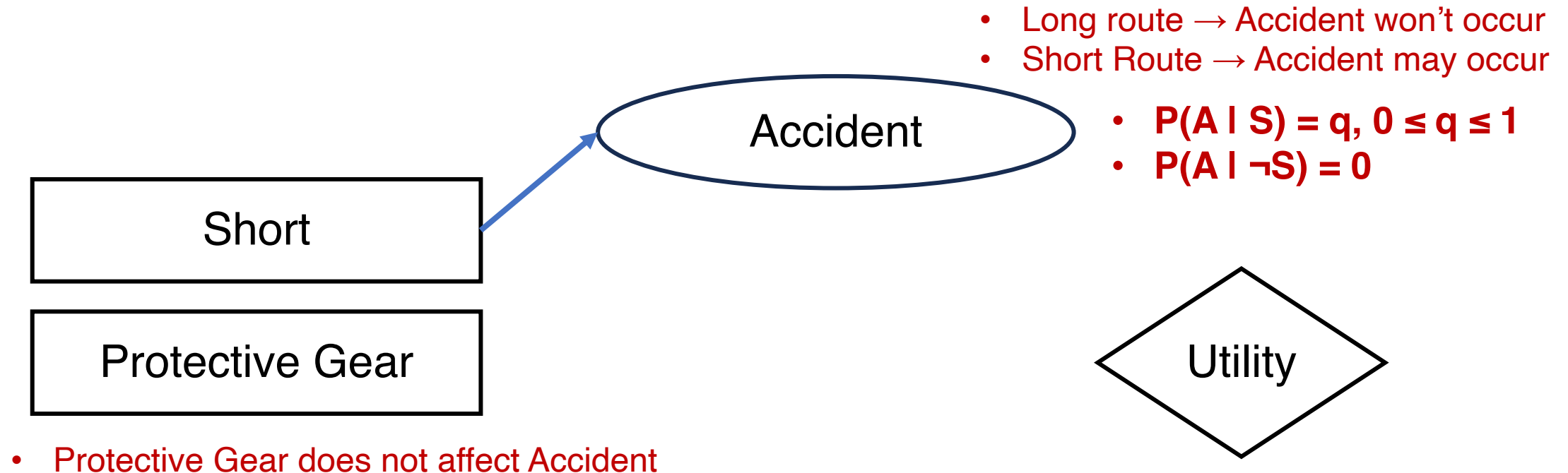
- Let's see what are the variables
- What are the random variables here?
  - A: whether an accident occurs or not
- What are the decision variables here (actions)?
  - S: whether the robot chooses the short route
  - P: whether the robot uses the protective gear

## Robot Decision Network



# Decision Theory and Networks

## Connecting the nodes in the Robot Decision Network



# Decision Theory and Networks

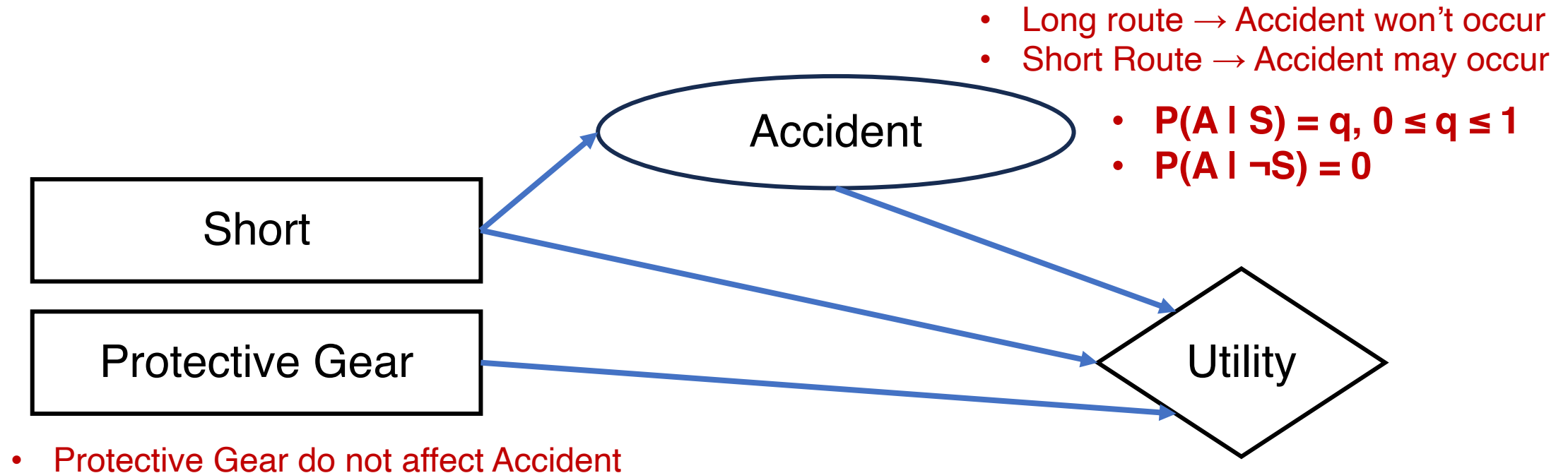
## **Let's talk about the robot's happiness**

- What variables do you think will affect/influence the robot's happiness?
  1. P only
  2. S only
  3. A only
  4. Two of (1), (2) and (3)
  5. All of (1), (2) and (3)



# Decision Theory and Networks

## Connecting the nodes in the Robot Decision Network



# Decision Theory and Networks

## A Critical Question

- When an accident does NOT happen, which of the following statements is true?
  1. The robot prefers not to wear the protective gear than to wear it.
  2. The robot prefers the long route over the short route
  3. Both (A) and (B) are true
  4. Both (A) and (B) are false

# Decision Theory and Networks

An example Utility function for the robot

| Variables                | State                     | $U(w_i)$ |
|--------------------------|---------------------------|----------|
| $\neg P, \neg S, \neg A$ | $w_0$ slow, no weight     | 6        |
| $\neg P, \neg S, A$      | $w_1$ impossible          |          |
| $\neg P, S, \neg A$      | $w_2$ quick, no weight    | 10       |
| $\neg P, S, A$           | $w_3$ severe damage       | 0        |
| $P, \neg S, \neg A$      | $w_4$ slow, extra weight  | 4        |
| $P, \neg S, A$           | $w_5$ impossible          |          |
| $P, S, \neg A$           | $w_6$ quick, extra weight | 8        |
| $P, S, A$                | $w_7$ moderate damage     | 2        |

# Decision Theory and Networks

**How does the robot's Utility/Happiness depend on the random variables and decision variables?**

- When an accident does NOT happen, does the robot prefer not to wear the protective gear or to wear it?
- When an accident does NOT happen, does the robot prefer the short route or the long route?

# Decision Theory and Networks

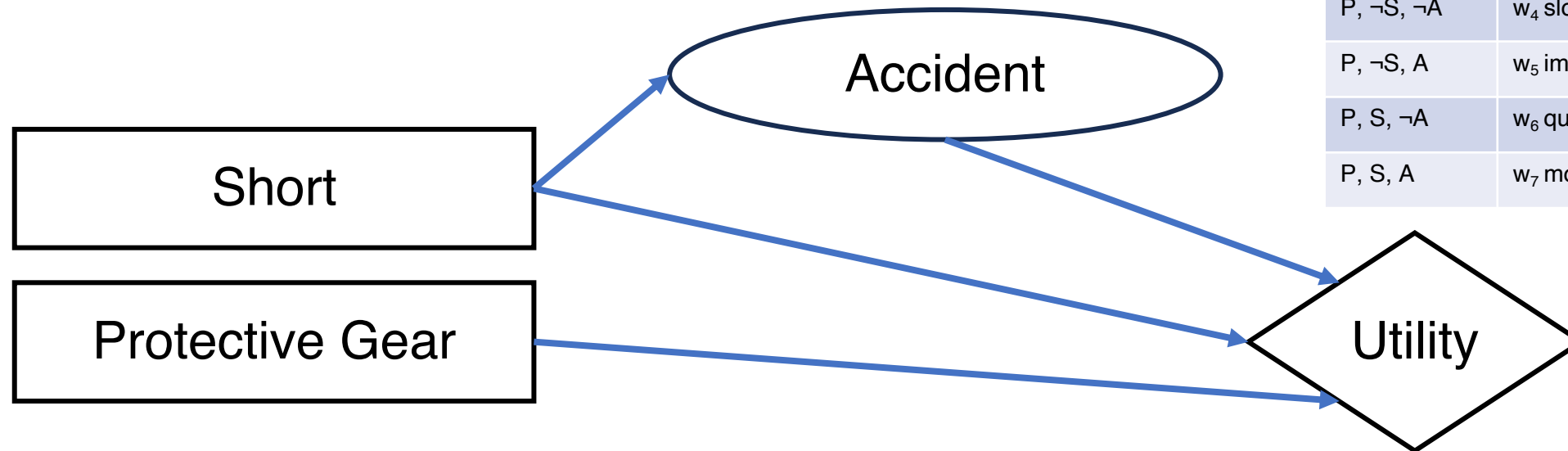
**How does the robot's Utility/Happiness depend on the random variables and decision variables?**

- When an accident does happen, does the robot prefer not to wear the protective gear or to wear it?
- When an accident does happen, does the robot prefer the short route or the long route?

# Decision Theory and Networks

## The complete Robot Decision Network

- $P(A \mid S) = q, 0 \leq q \leq 1$
- $P(A \mid \neg S) = 0$



| Variables                | State                     | $U(w_i)$ |
|--------------------------|---------------------------|----------|
| $\neg P, \neg S, \neg A$ | $w_0$ slow, no weight     | 6        |
| $\neg P, \neg S, A$      | $w_1$ impossible          |          |
| $\neg P, S, \neg A$      | $w_2$ quick, no weight    | 10       |
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| $P, \neg S, A$           | $w_5$ impossible          |          |
| $P, S, \neg A$           | $w_6$ quick, extra weight | 8        |
| $P, S, A$                | $w_7$ moderate damage     | 2        |

# Decision Theory and Networks

## Evaluating the Robot Decision Network

- How do we choose an action?
  - Set evidence variables for the current state
  - For each possible value of the decision node
    - Set decision node to that value
    - Calculate the posterior probabilities for the parent nodes of the utility node
    - Calculate expected utility for the action
  - Return action with the highest expected utility

# Decision Theory and Networks

## Evaluating the Robot Decision Network

- What is the agent's utility of not wearing protective gear and choosing the long route?



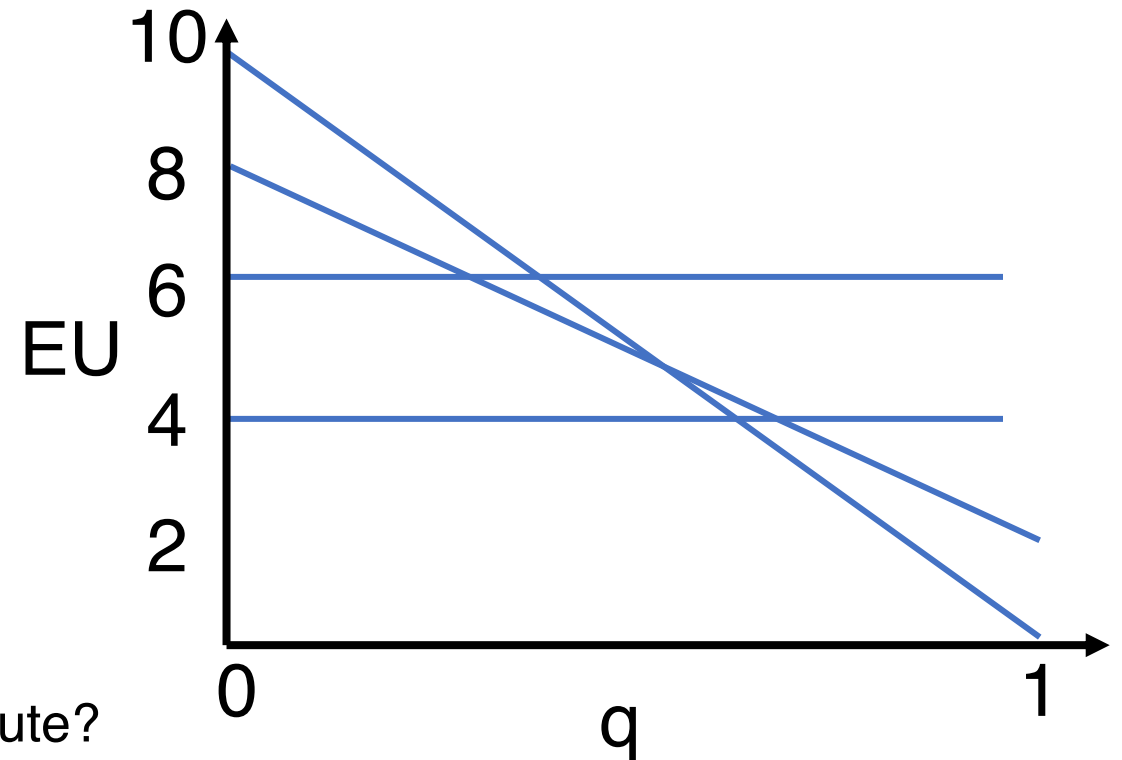
# Decision Theory and Networks

## Evaluating the Robot Decision Network

- Similarly evaluating this for the other three cases gives us the following:

- $EU(\neg P, \neg S) = 6$
- $EU(\neg P, S) = 10 - 10q$
- $EU(P, \neg S) = 4$
- $EU(P, S) = 8 - 6q$

- Should it wear protective gear or not?
- Should it choose the long or the short route?



# Decision Theory and Networks

## Evaluating the Robot Decision Network

- Similarly evaluating this for the other three cases gives us the following:

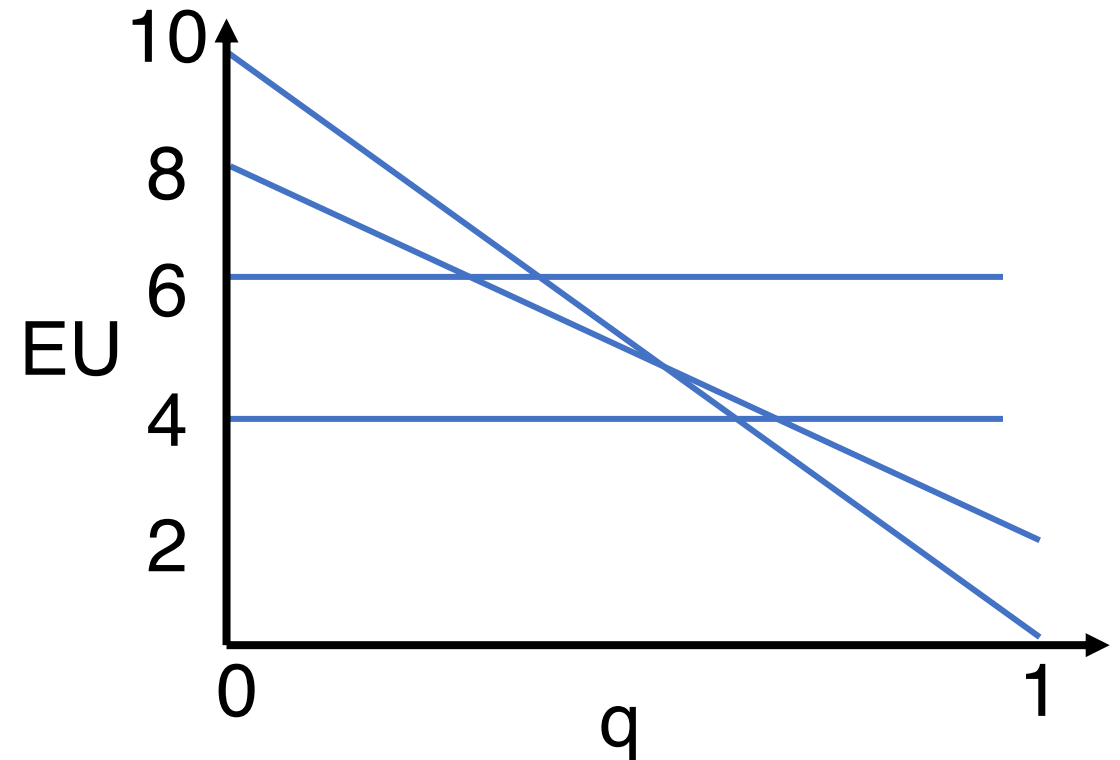
- $EU(\neg P, \neg S) = 6$

- $EU(\neg P, S) = 10 - 10q$

- $EU(P, \neg S) = 4$

- $EU(P, S) = 8 - 6q$

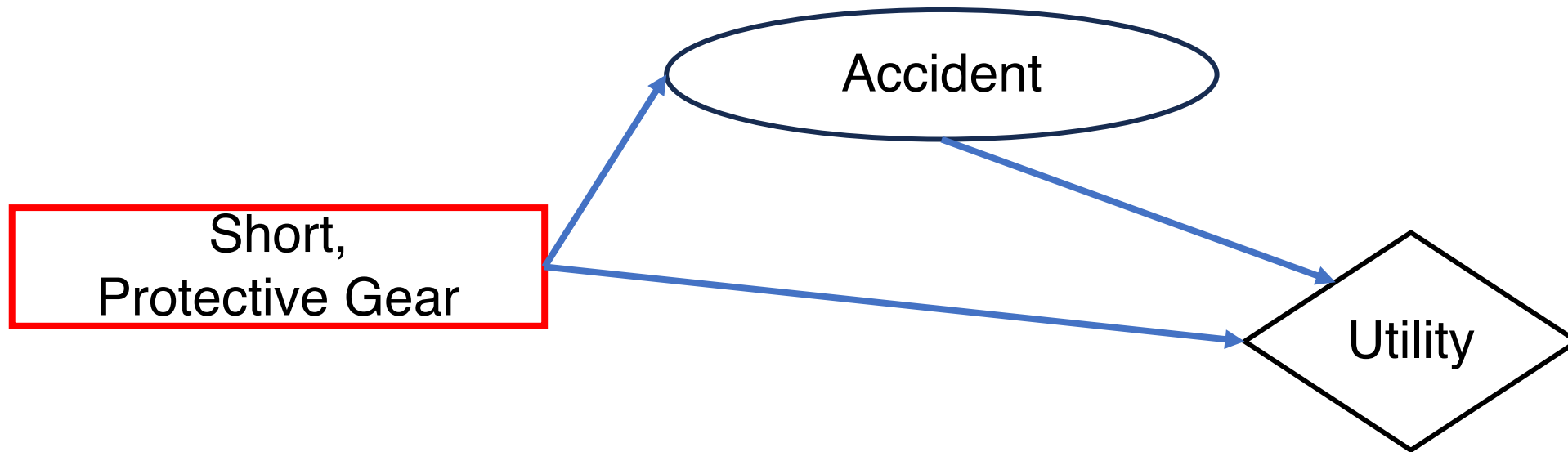
- Idea of a policy and an optimal policy

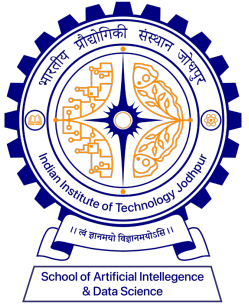


# Decision Theory and Networks

## Evaluating the Robot Decision Network

- Also possible to solve this using the variable elimination algorithm





# Advanced Artificial Intelligence

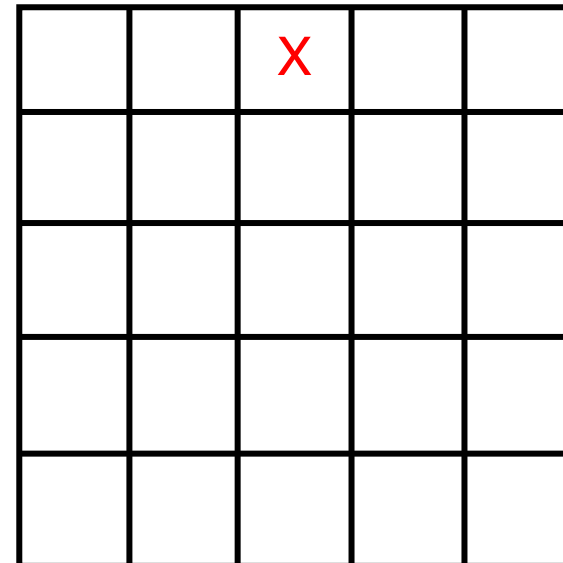
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# Decision Theory and Networks

## Value of Information

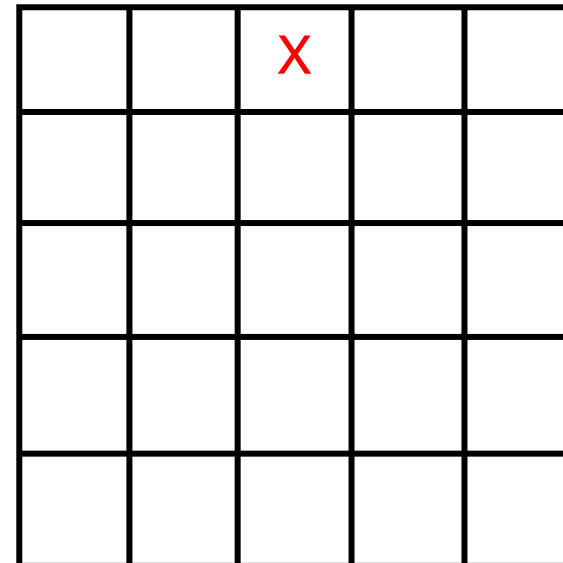
- Most important part of decision-making is knowing what question to ask.
- Information value theory enables agent to choose what information to acquire.
- Take this example for instance:
  - $n$  blocks of land
  - 1 of them contains oil worth  $C$  dollars
  - Each block cost –  $C/n$



# Decision Theory and Networks

## Value of Information

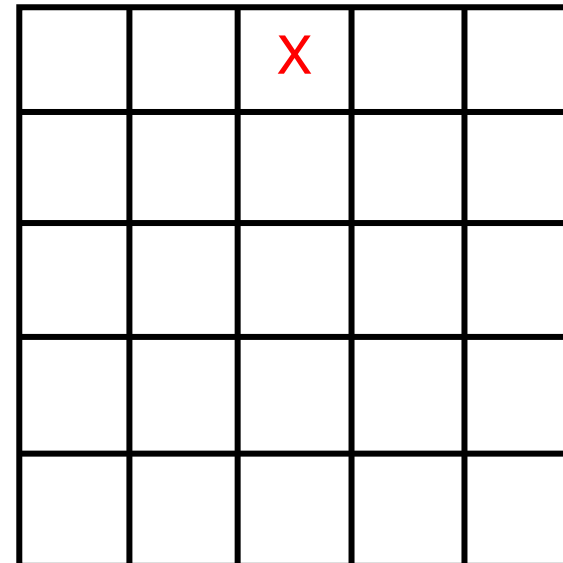
- Most important part of decision-making is knowing what question to ask.
- Information value theory enables agent to choose what information to acquire.
- Take this example for instance:
  - Seismologist gives clue
  - Company buys it for  $C/n$  dollars
  - In case block has **oil**:
    - Profit:  $C - C/n = (n-1)C/n$



# Decision Theory and Networks

## Value of Information

- Most important part of decision-making is knowing what question to ask.
- Information value theory enables agent to choose what information to acquire.
- Take this example for instance:
  - Seismologist gives clue
  - Company buys it for  $C/n$  dollars
  - In case block has **no oil**:
    - Profit:  $C/(n-1) - C/n = C/n(n-1)$



# Decision Theory and Networks

## Value of Information

- Most important part of decision-making is knowing what question to ask.
- Information value theory enables agent to choose what information to acquire.
- Take this example for instance:
  - Total expected profit with information

$$\begin{aligned} \text{Expected Profit} &= \frac{1}{n} \cdot \frac{(n-1)C}{n} + \frac{n-1}{n} \cdot \frac{C}{n(n-1)} \\ &= \frac{C}{n} \end{aligned}$$

|  |  |   |  |  |
|--|--|---|--|--|
|  |  | X |  |  |
|  |  |   |  |  |
|  |  |   |  |  |
|  |  |   |  |  |
|  |  |   |  |  |



# Decision Theory and Networks

## Value of Perfect Information (VPI)

- Let agent's current evidence be  $\mathbf{e}$ .

$$EU(\alpha|\mathbf{e}) = \max_a \sum_{s'} P(\text{RESULT}(a) = s' | a, \mathbf{e}) U(s')$$

- After new evidence  $e_j$  is obtained,

$$EU(\alpha_{e_j}|\mathbf{e}, e_j) = \max_a \sum_{s'} P(\text{RESULT}(a) = s' | a, \mathbf{e}, e_j) U(s')$$

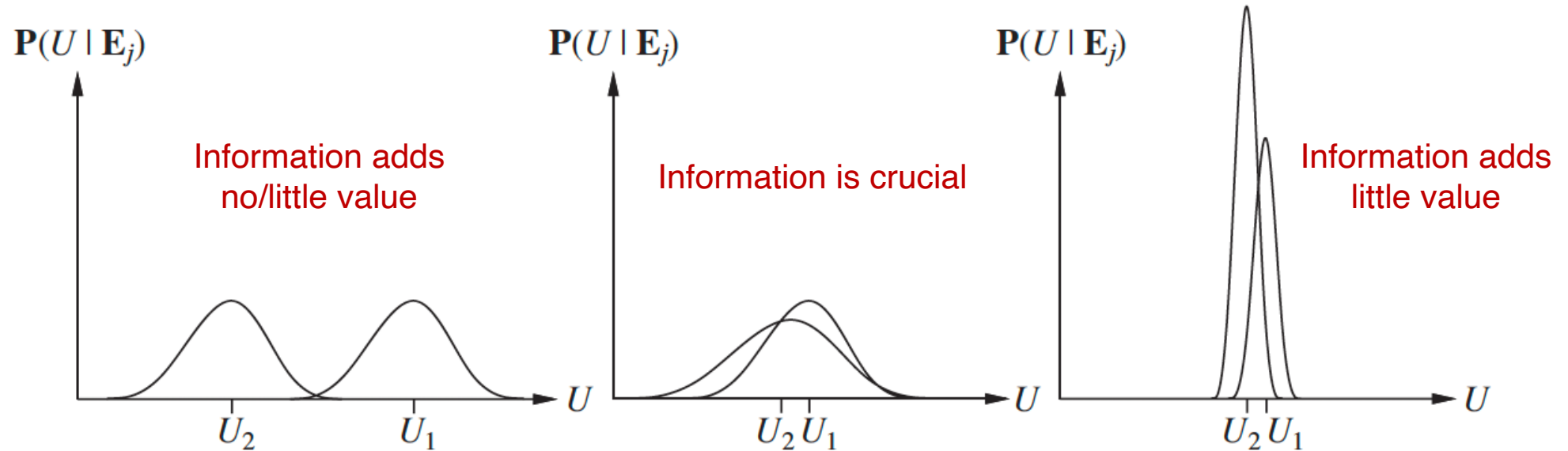
- $E_j$  is a random variable so the value of obtaining that information is

$$VPI_{\mathbf{e}}(E_j) = \left( \sum_k P(E_j = e_{jk}|\mathbf{e}) EU(\alpha_{e_{jk}}|\mathbf{e}, E_j = e_{jk}) \right) - EU(\alpha|\mathbf{e})$$

# Decision Theory and Networks

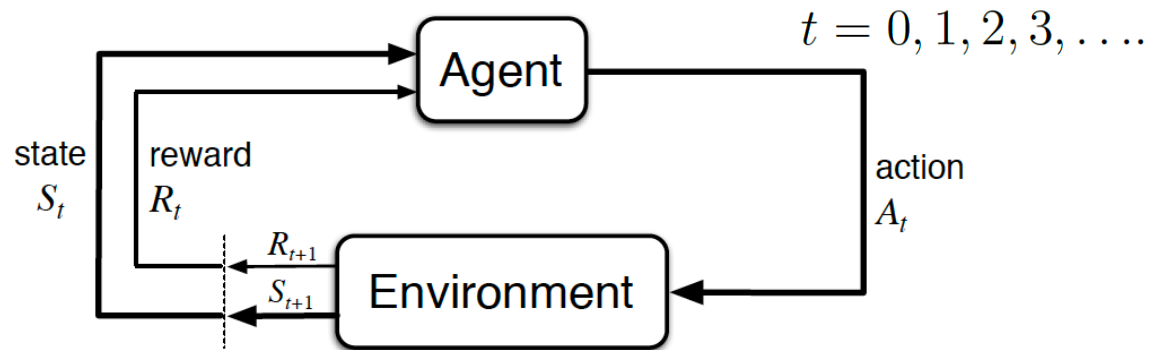
## Value of Perfect Information (VPI)

- More intuitively it means the following, consider only 2 actions with utilities  $U_1$  and  $U_2$



# Reinforcement Learning – the problem at hand

The interaction between agent and environment



Goal of RL: Agent to MAXIMIZE rewards

\*Tension between breadth of applicability and mathematical tractability

Components of RL

- Agent and Environment
- Feedback: Rewards  $R_t \in \mathbb{R}$ .
- States  $S_t \in \mathcal{S}$
- Actions  $A_t \in \mathcal{A}(S_t)$
- Policy  $\pi_t(a|s)$

# Scenarios where RL is applicable

Controlling a bioreactor



A cleaning robot



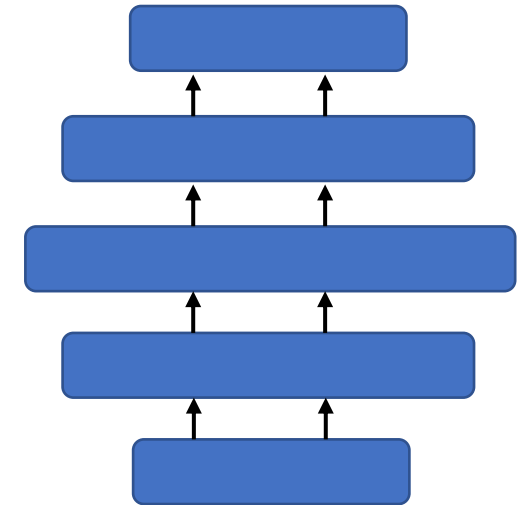
Game Engines: Chess/Alpha zero



Human behavior



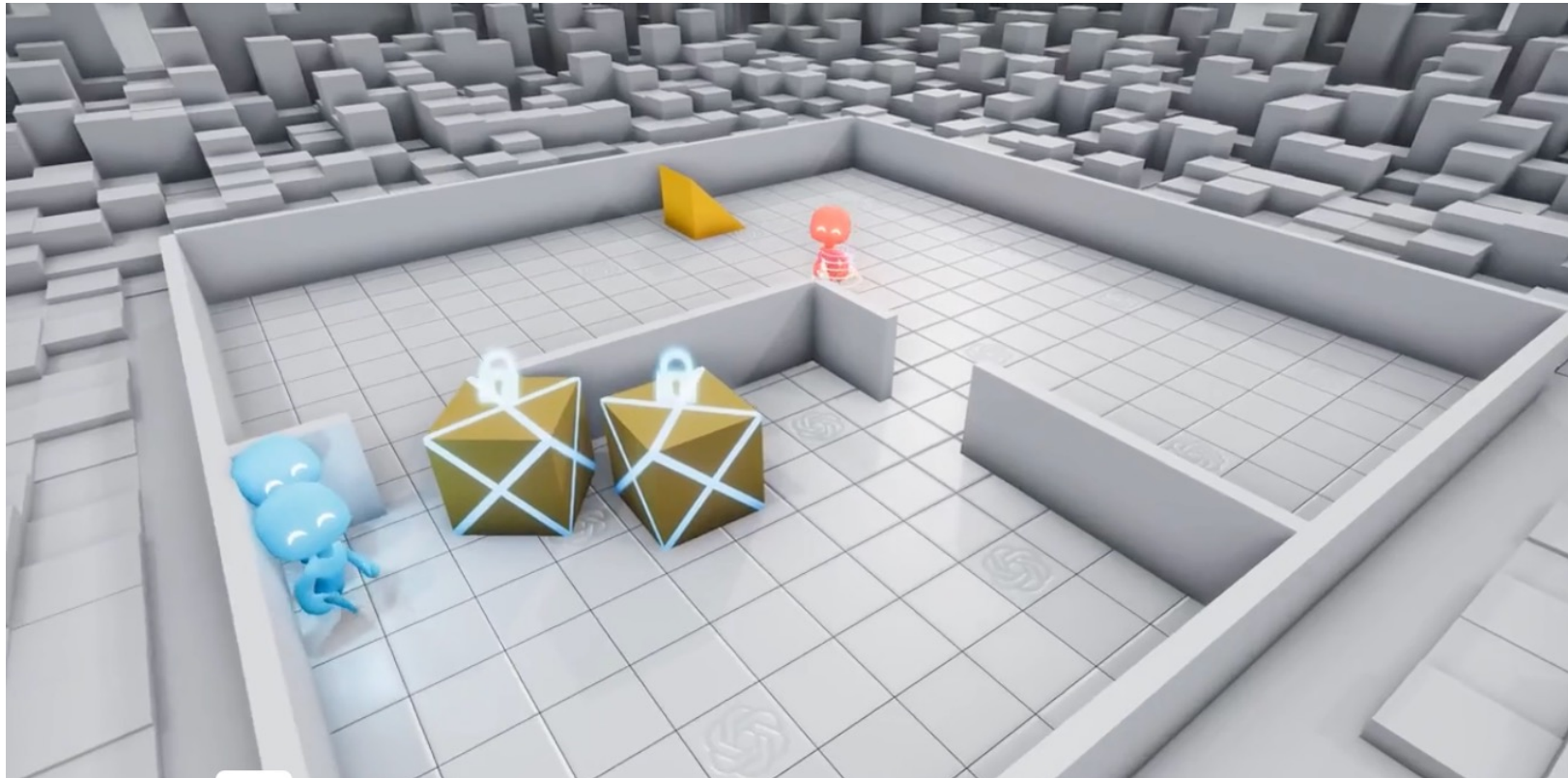
Large Parameter Models



Can you all think of examples?

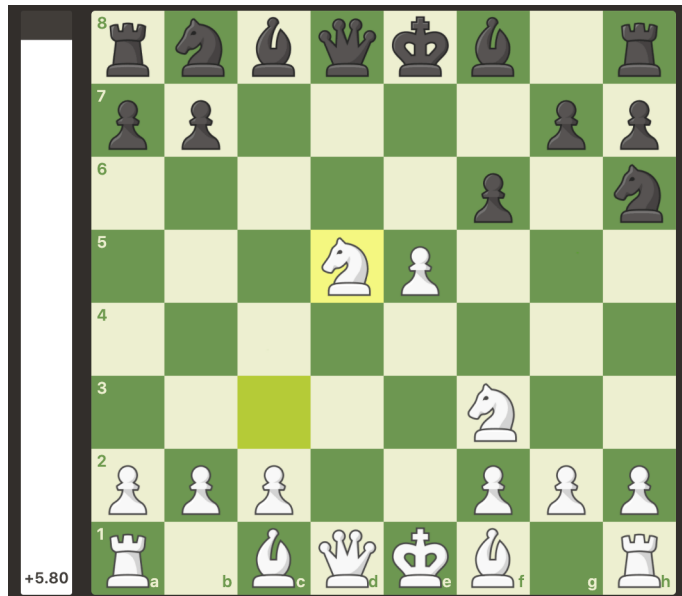
# A fascinating example of RL: Emergent behavior

<https://openai.com/research/emergent-tool-use>



# Goals and Returns: Maximizing expected rewards

Distinctive feature of RL: Use of a reward signal to formalize the goal



Place the line not at the limit of the body but  
at the limit of control

The line between agent and environment

- Environment provides the rewards
- Robot learning: for internal battery storage
- Robot learning: limbs to interact in the external world
- The driving problem: Body meets machine, rubber meets road, brain meets body

# Returns: Maximizing expected rewards

Sequence of rewards:  $R_{t+1}, R_{t+2}, R_{t+3}, \dots,$

Cumulative return:  $G_t = R_{t+1} + R_{t+2} + R_{t+3} + \dots + R_T$

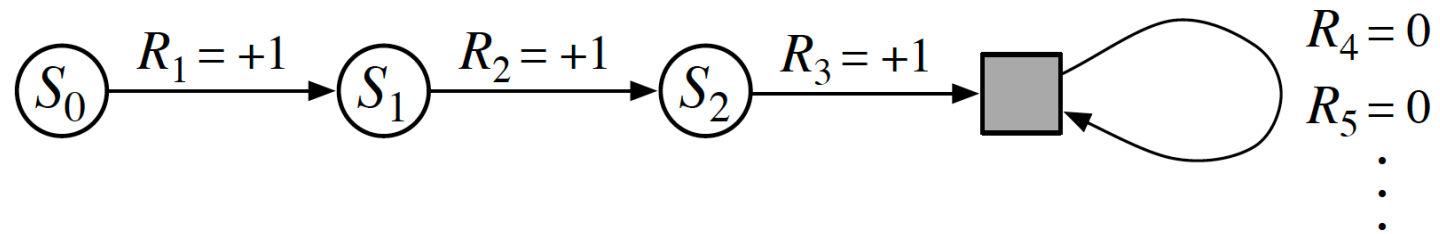
But if time goes on till infinity,  $T \rightarrow \infty$

Discounted return:  $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$

where  $\gamma$  is the discount factor/rate, s.t.  $0 \leq \gamma < 1$

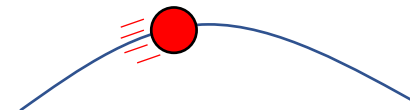
# Markov Property

Formalizing the episodic process of state-action-reward transitions



The Idea of a Markov Property: **Current state retains all the information from the past**

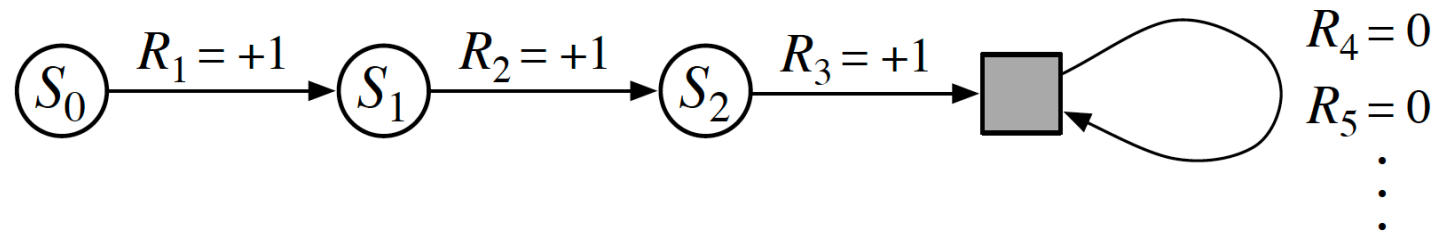
Independence of path





# Markov Property

Formalizing the episodic process of state-action-reward transitions



Formalizing the Markov property

$$\Pr\{R_{t+1} = r, S_{t+1} = s' \mid S_0, A_0, R_1, \dots, S_{t-1}, A_{t-1}, R_t, S_t, A_t\} \quad \text{Not Markovian}$$

$$p(s', r | s, a) = \Pr\{R_{t+1} = r, S_{t+1} = s' \mid S_t, A_t\} \longrightarrow \text{Predict the next state } s' \text{ and reward } r, \text{ based on the current state-action pair}$$

$\forall s', r, S_t \text{ and } A_t$



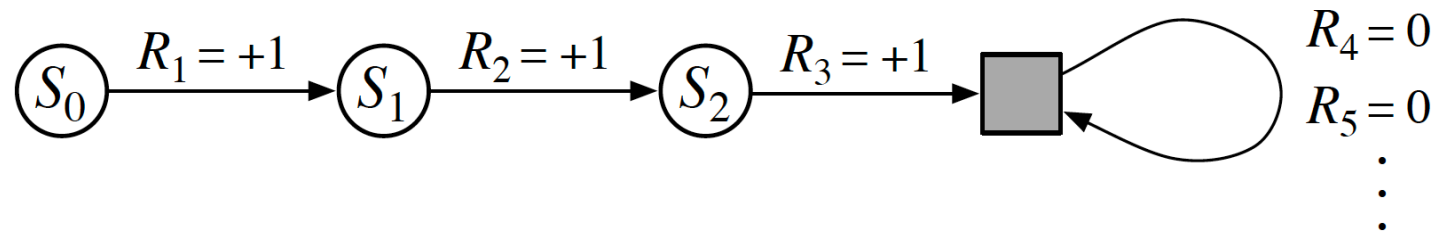
# Advanced Artificial Intelligence

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# Finite Markov Decision Processes (MDPs)

Formalizing the episodic process of state-action-reward transitions



Dynamics of a Finite MDP

$$p(s', r | s, a) = \Pr\{S_{t+1} = s', R_{t+1} = r \mid S_t = s, A_t = a\}$$

State transition probability

$$p(s' | s, a) = \Pr\{S_{t+1} = s' \mid S_t = s, A_t = a\} = \sum_{r \in \mathcal{R}} p(s', r | s, a)$$

Expected Return of a Finite MDP

$$r(s, a) = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a] = \sum_{r \in \mathcal{R}} r \sum_{s' \in \mathcal{S}} p(s', r | s, a)$$

State-action-state triple

$$r(s, a, s') = \mathbb{E}[R_{t+1} \mid S_t = s, A_t = a, S_{t+1} = s'] = \frac{\sum_{r \in \mathcal{R}} r p(s', r | s, a)}{p(s' | s, a)}$$

# Finite Markov Decision Processes (MDPs)

A cleaning robot



State Space

$$\mathcal{S} = \{\text{high}, \text{low}\}$$

Action Space

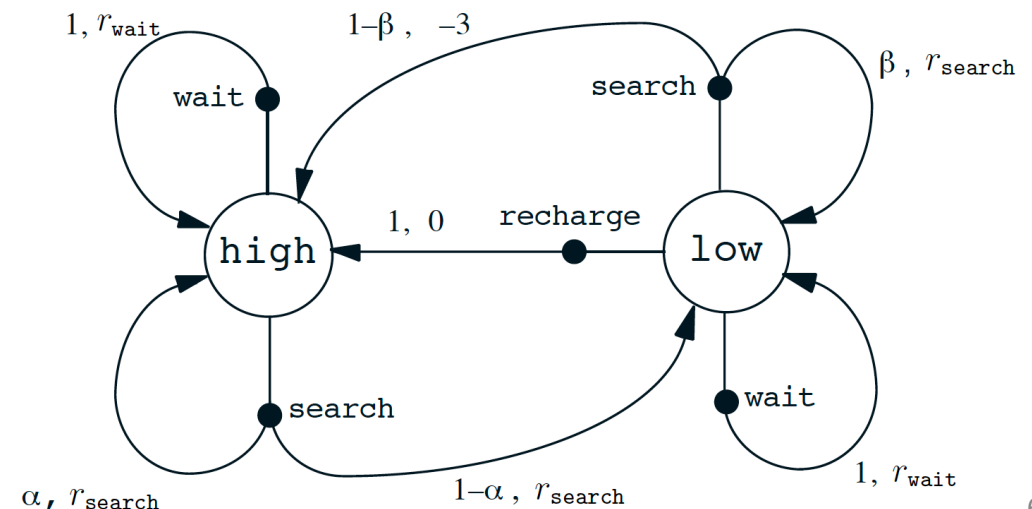
$$\mathcal{A}(\text{high}) = \{\text{search}, \text{wait}\}$$

$$\mathcal{A}(\text{low}) = \{\text{search}, \text{wait}, \text{recharge}\}$$

Transition Matrix

| $s$  | $s'$ | $a$      | $p(s' s, a)$ | $r(s, a, s')$       |
|------|------|----------|--------------|---------------------|
| high | high | search   | $\alpha$     | $r_{\text{search}}$ |
| high | low  | search   | $1 - \alpha$ | $r_{\text{search}}$ |
| low  | high | search   | $1 - \beta$  | $-3$                |
| low  | low  | search   | $\beta$      | $r_{\text{search}}$ |
| high | high | wait     | 1            | $r_{\text{wait}}$   |
| high | low  | wait     | 0            | $r_{\text{wait}}$   |
| low  | high | wait     | 0            | $r_{\text{wait}}$   |
| low  | low  | wait     | 1            | $r_{\text{wait}}$   |
| low  | high | recharge | 1            | 0                   |
| low  | low  | recharge | 0            | 0                   |

Transition Graph



# Finite MDPs Question

- Brainstorm tasks that can be fit into the RL framework: clearly stating the state, actions and reward distributions

# Value Functions

How good is it for the agent to be in that specific state?

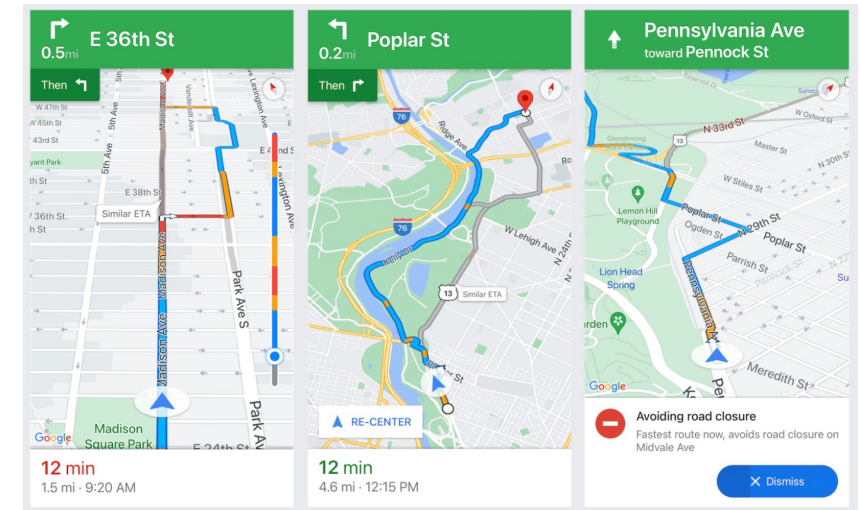
The State-value function under policy  $\pi$

$$v_{\pi}(s) = \mathbb{E}_{\pi}[G_t \mid S_t = s] = \mathbb{E}_{\pi} \left[ \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s \right]$$

The Action-value function under policy  $\pi$

$$q_{\pi}(s, a) = \mathbb{E}_{\pi}[G_t \mid S_t = s, A_t = a] = \mathbb{E}_{\pi} \left[ \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \mid S_t = s, A_t = a \right]$$

## Navigation Problem



# Value Functions: The Bellman Equation

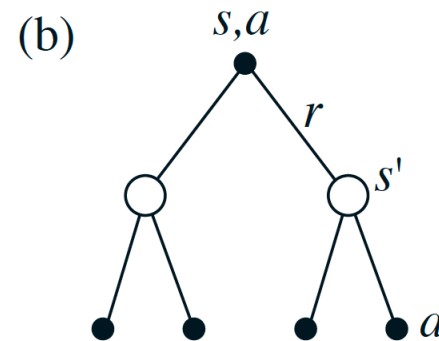
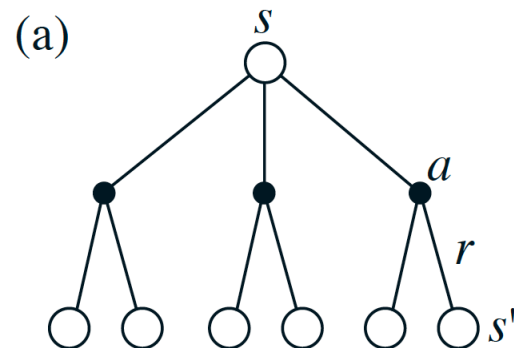
Demonstrate relationship between value of current state and the successor state

# Value Functions: The Bellman Equation

Demonstrate relationship between value of current state and the successor state

$$v_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) \left[ r + \gamma v_{\pi}(s') \right]$$

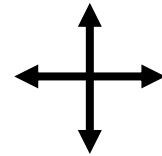
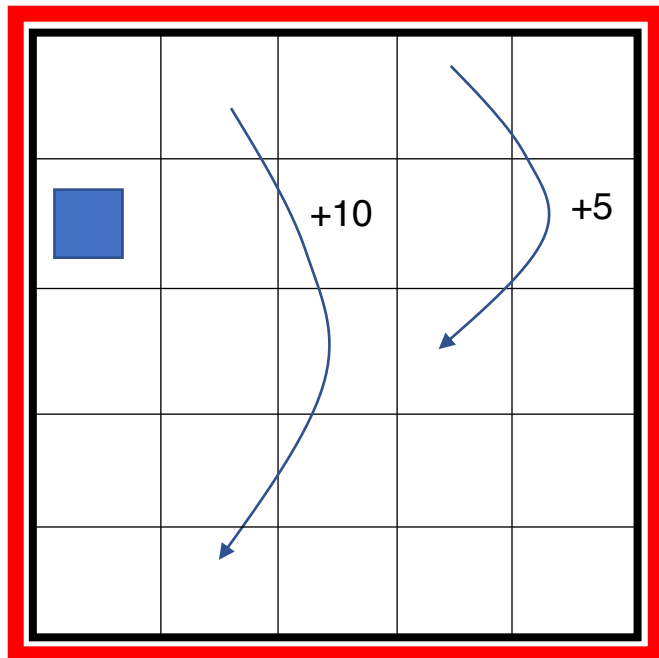
Backup Diagrams





# Value Functions – Gridworld

Discrete movements in 2D space – with defined actions



Actions

Policy: Equal Probability to move in each direction

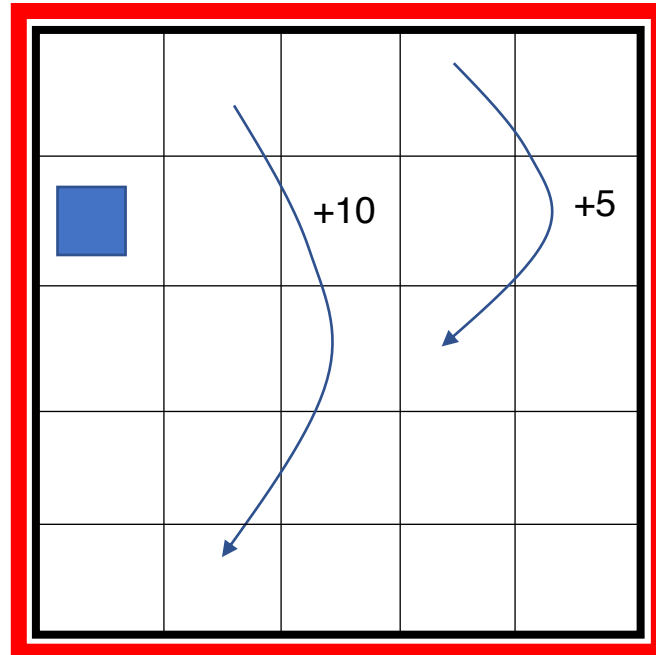
Value function formed

|      |      |      |      |      |
|------|------|------|------|------|
| 3.3  | 8.8  | 4.4  | 5.3  | 1.5  |
| 1.5  | 3.0  | 2.3  | 1.9  | 0.5  |
| 0.1  | 0.7  | 0.7  | 0.4  | -0.4 |
| -1.0 | -0.4 | -0.4 | -0.6 | -1.2 |
| -1.9 | -1.3 | -1.2 | -1.4 | -2.0 |

# Ideas of optimal policy and value function

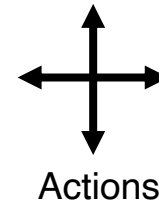
Optimal Policy:  $\pi_*$

Optimal State-value function  $v_*(s) = \max_{\pi} v_{\pi}(s)$



Value function formed

|      |      |      |      |      |
|------|------|------|------|------|
| 3.3  | 8.8  | 4.4  | 5.3  | 1.5  |
| 1.5  | 3.0  | 2.3  | 1.9  | 0.5  |
| 0.1  | 0.7  | 0.7  | 0.4  | -0.4 |
| -1.0 | -0.4 | -0.4 | -0.6 | -1.2 |
| -1.9 | -1.3 | -1.2 | -1.4 | -2.0 |



$v_*$

|      |      |      |      |      |
|------|------|------|------|------|
| 22.0 | 24.4 | 22.0 | 19.4 | 17.5 |
| 19.8 | 22.0 | 19.8 | 17.8 | 16.0 |
| 17.8 | 19.8 | 17.8 | 16.0 | 14.4 |
| 16.0 | 17.8 | 16.0 | 14.4 | 13.0 |
| 14.4 | 16.0 | 14.4 | 13.0 | 11.7 |

$\pi_*$

|   |   |   |   |   |
|---|---|---|---|---|
| → | ↕ | ← | ↕ | ← |
| ↙ | ↑ | ↘ | ← | ← |
| ↙ | ↑ | ↘ | ↘ | ↘ |
| ↙ | ↑ | ↘ | ↘ | ↘ |
| ↙ | ↑ | ↘ | ↘ | ↘ |

# What do we need to compute Value?

Let's look at the Bellman Equation

$$v_{\pi}(s) = \sum_a \pi(a|s) \sum_{s', r} p(s', r|s, a) \left[ r + \gamma v_{\pi}(s') \right]$$

# Model-based learning

Most of the times we don't know how the environment works  $P(s'|s, a)$

$$R(s, a, s')$$

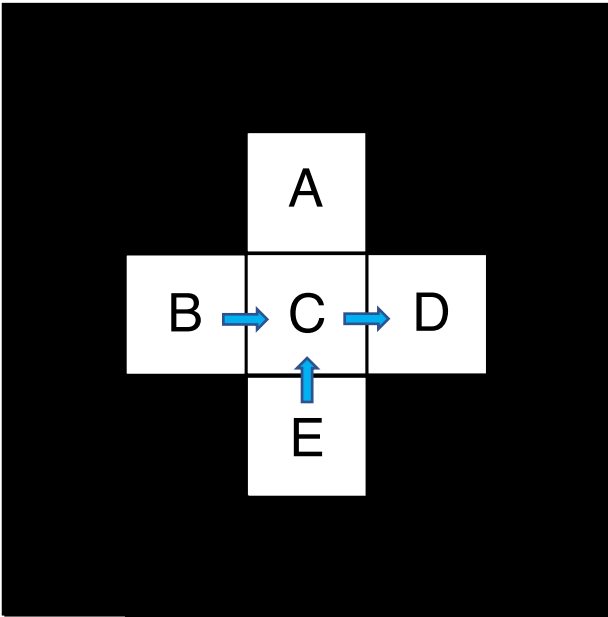
Learn an approximate model based on experience

Have to make an action and then know

Solve for the value function, i.e., the Bellman Equation

# Model-based learning

Fixed Policy,  $\pi$



Take  $\gamma = 1$

Observed Transitions

## Episode-1

B, east, C, -1  
C, east, D, -1  
D, exit, x, +10

## Episode-2

B, east, C, -1  
C, east, D, -1  
D, exit, x, +10

## Episode-3

E, north, C, -1  
C, east, D, -1  
D, exit, x, +10

## Episode-4

E, north, C, -1  
C, east, A, -1  
A, exit, x, -10

# Model-free learning

An Example: Age, we know  $P(A)$

$$E[A] = \sum_a P(A) \cdot a$$

Model-based: Come up with an approximation for  $P(A)$ , collect samples  $a_1, a_2, \dots, a_N$

$$\hat{P}(A) = \frac{\text{num}(a)}{N}$$

$$E[A] \approx \sum_a \hat{P}(A) \cdot a$$

Model-free: Directly compute it from the samples

$$E[A] \approx \frac{1}{N} \sum_i a_i$$



# Advanced Artificial Intelligence

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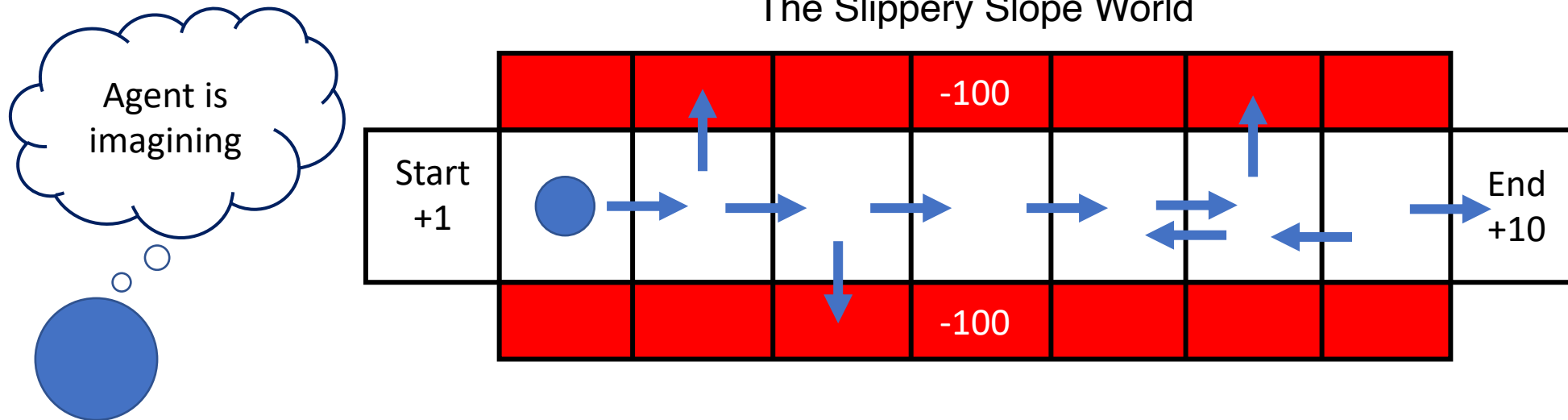
Email: [vigneshmdharan@iitj.ac.in](mailto:vigneshmdharan@iitj.ac.in)

# Types of Reinforcement Learning

|                           | Passive RL                                                | Active RL                                                 |
|---------------------------|-----------------------------------------------------------|-----------------------------------------------------------|
| <b>States and Rewards</b> | Observe only visited states and rewards associated        | Observe only visited states and rewards associated        |
| <b>Transitions</b>        | Observe only transitions occurring from performed actions | Observe only transitions occurring from performed actions |
| <b>Policy/Decisions</b>   | No learning/updating policy                               | Learning/Updating policy or decisions                     |
| <b>Goal</b>               | <b>Learn State-values</b>                                 | <b>Learn Action-values</b>                                |



# Passive Reinforcement Learning

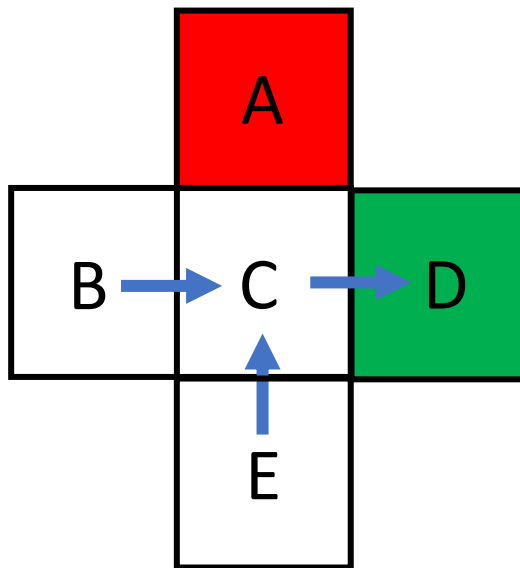


- Don't know the transitions or reward probabilities.
- Simulating the actions in the world.
- **Ultimate Goal: Learn optimal values**

# Passive RL: Direct Evaluation

Average all the observed sample cumulative returns from that state

The Slippery Slope World  
Fixed Policy,  $\pi$



Let's take  $\gamma = 1$

Observed Transitions

## Episode-1

B, east, C, -1  
C, east, D, -1  
D, exit, x, +10

## Episode-2

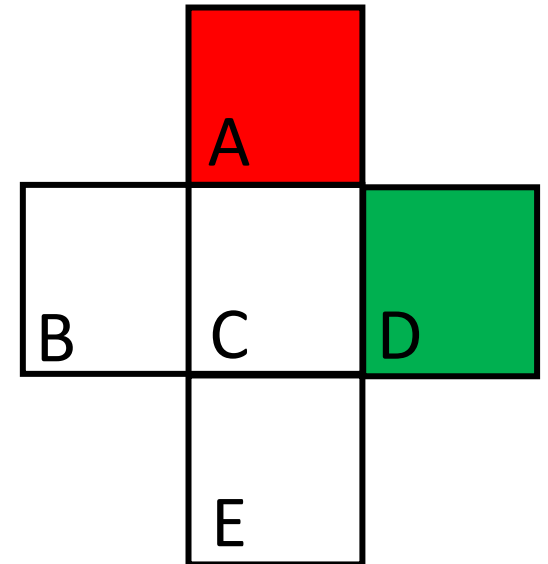
B, east, C, -1  
C, east, D, -1  
D, exit, x, +10

## Episode-3

E, north, C, -1  
C, east, D, -1  
D, exit, x, +10

## Episode-4

E, north, C, -1  
C, east, A, -1  
A, exit, x, -10



# Passive RL: Sample based policy evaluation

Improving our estimate of  $V$  using the following:

$$v_{k+1}^{\pi}(s) \leftarrow \sum_{s'} P(s'|s, \pi(s)) [R(s, \pi(s), s') + \gamma v_k^{\pi}(s')]$$

Take samples of outcome  $s'$  and average:

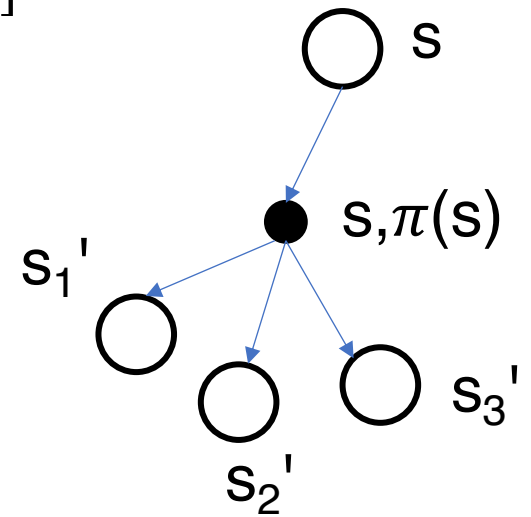
$$sample_1 = [R(s, \pi(s), s'_1) + \gamma v^{\pi} s'_1]$$

$$sample_2 = [R(s, \pi(s), s'_2) + \gamma v^{\pi} s'_2]$$

...

$$sample_N = [R(s, \pi(s), s'_N) + \gamma v^{\pi} s'_N]$$

$$v_{k+1}^{\pi}(s) \leftarrow \frac{1}{N} \sum_{i=1}^N sample_i$$



Issues with this approach

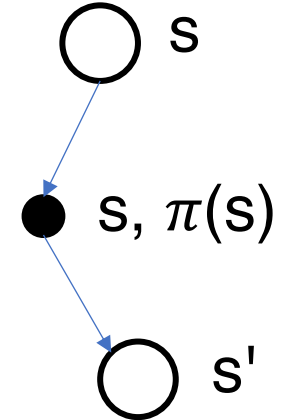
# Passive RL: Temporal Difference Learning

## Main Intuition:

- Learning the value functions from every transition the agent makes in the environment.
- The likelier next states,  $s'$ , will contribute to the value more.

## TD Learning of values:

- Policy is fixed still; we are learning the values  $V^\pi(s)$ .
- More samples of the value of the likelier next states will contribute to the value of the current state; running average



# Passive RL: Temporal Difference Learning

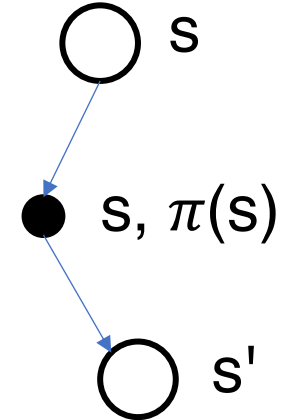
## Main Intuition:

- Learning the value functions from every transition the agent makes in the environment.
- The likelier next states,  $s'$ , will contribute to the value more.

Sample Return:  $\text{sample} = R(s, \pi(s), s') + \gamma V^\pi(s')$

Update to  $V(s)$ :  $V^\pi(s) \leftarrow (1 - \alpha)V^\pi(s) + \alpha(\text{sample})$

$$V^\pi(s) \leftarrow V^\pi(s) + \alpha(\text{sample} - V^\pi(s))$$



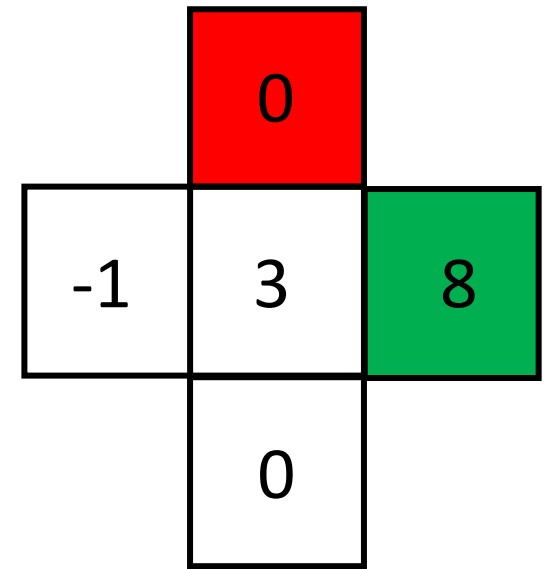
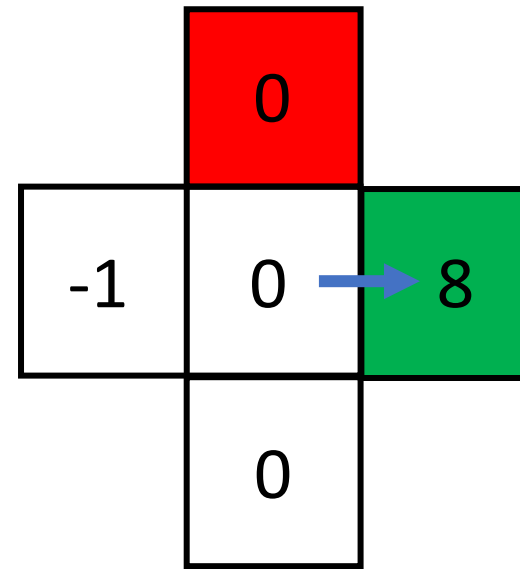
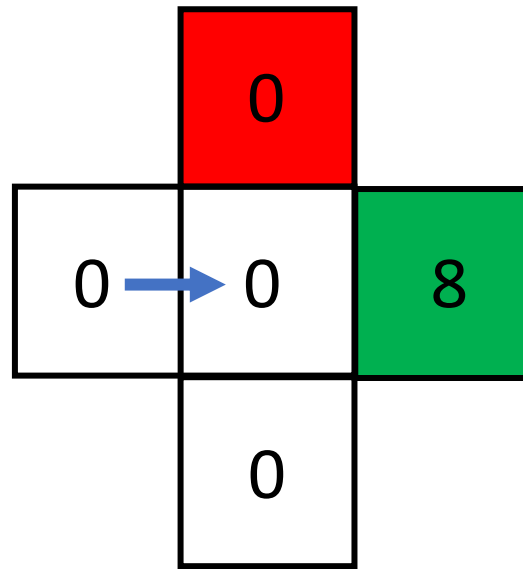
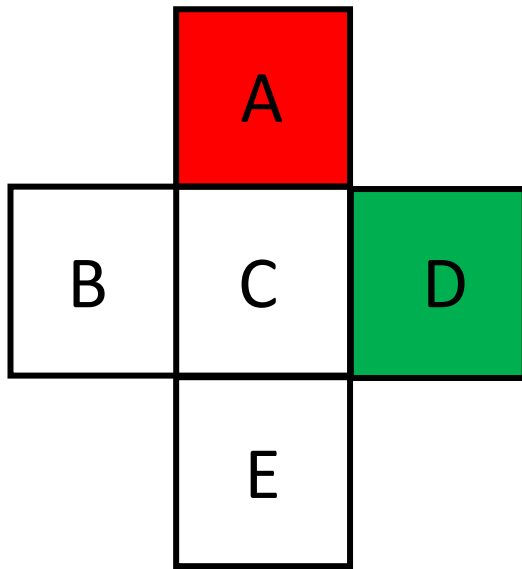
# Passive RL: Temporal Difference Learning

The Slippery Slope World  
Fixed Policy,  $\pi$

Observed Transition

B, east, C, -2

C, east, D, -2



Let's take  $\gamma = 1$ ,  $\alpha = 0.5$

$$V^\pi(s) \leftarrow V^\pi(s) + \alpha(R(s, \pi(s), s') + \gamma V^\pi(s') - V^\pi(s))$$

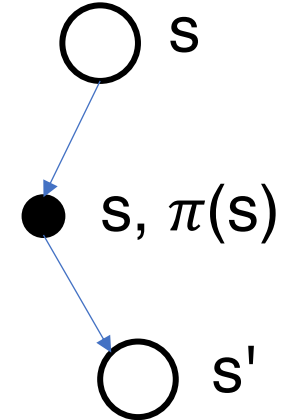
# Passive RL: TD Learning Problems

Model-free way of performing policy evaluation, mimicking the Bellman Equation with moving averages

Still, we are stuck with the policy we started with and can't find anything better

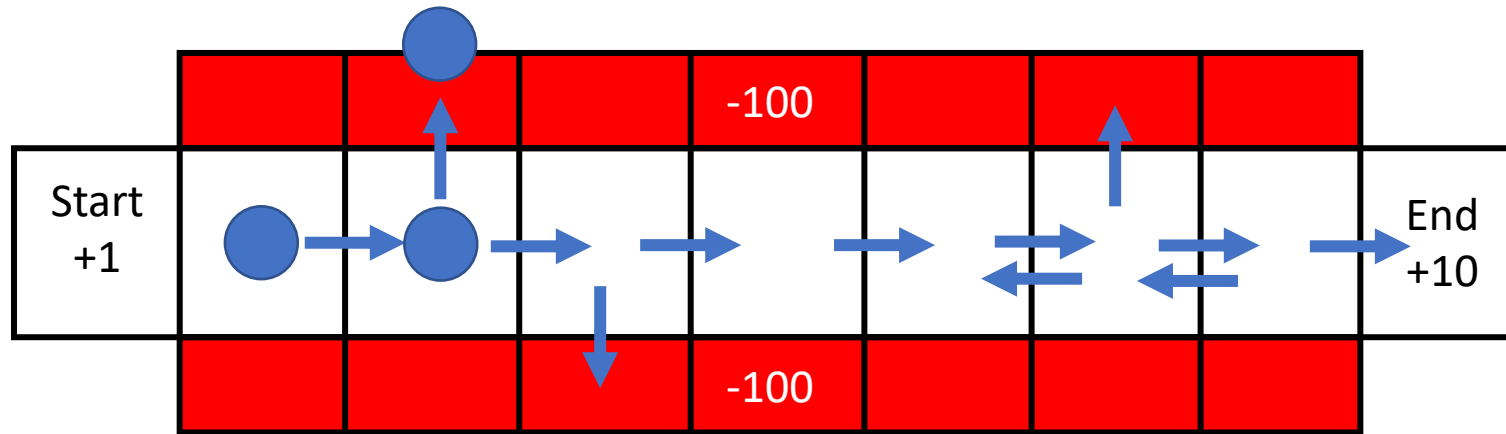
$$V_{\pi}(s)$$

$$Q(s,a)$$



# Active Reinforcement Learning

The Slippery Slope World



## Full Reinforcement Learning Problem

- Don't know the transitions or reward probabilities.
- Perform actions in the world.
- **Ultimate Goal: Learn optimal values and policy**

Q-values, aka action values,  
instead of state values,  $V$ .



# Active RL: Q-Learning

Sample-based Q-value iteration

$$Q_{k+1}(s, a) \leftarrow \sum_{s'} P(s'|s, a) [R(s, a, s') + \gamma \max_{a'} Q_k(s', a')]$$

Receive a sample: (s, a, s', r)

The old estimate:  $Q(s, a)$

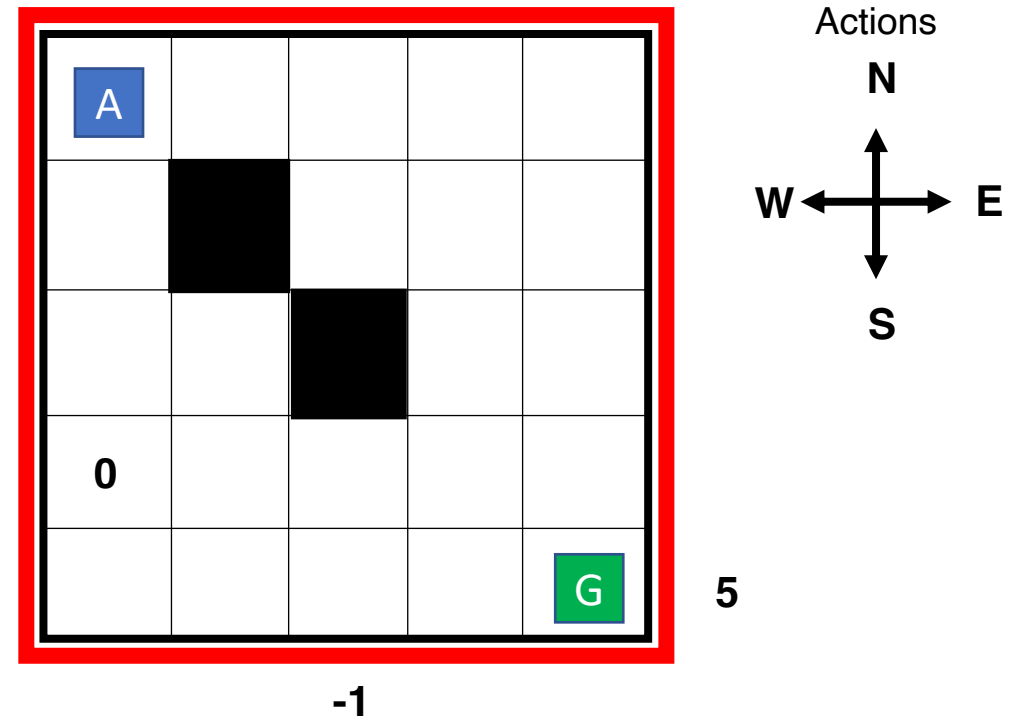
The new sample estimate:  $\text{sample} = [R(s, a, s') + \gamma \max_{a'} Q(s', a')]$

**Incorporate the new estimate in the running average:**

$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha(\text{sample})$$

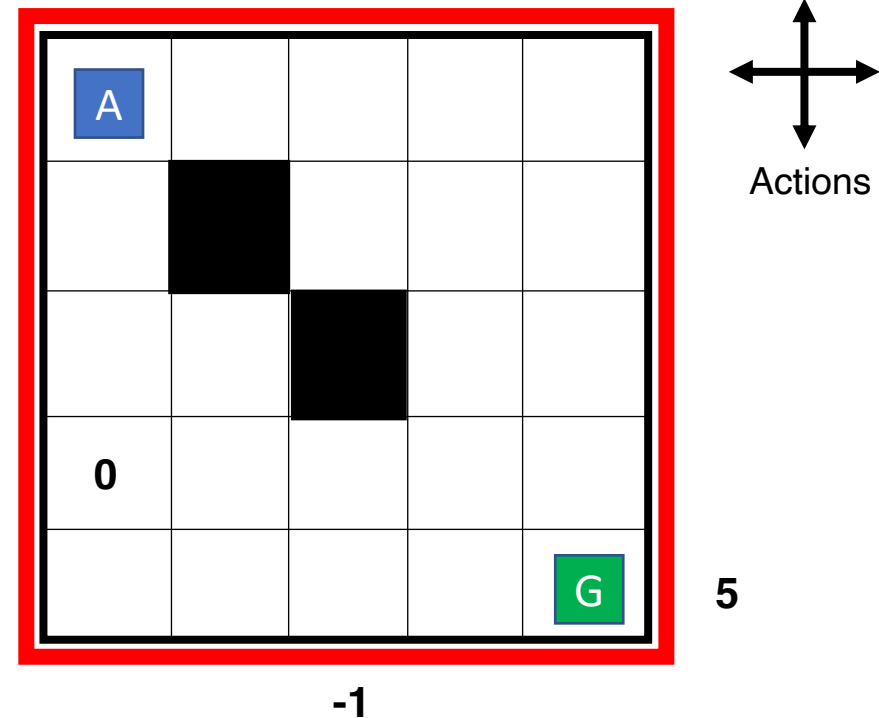
# Gridworld – Q Learning

- Create a 5x5 grid world
- An agent to move around
- Four possible actions
- A goal state.
- Reward at Goal = 5
- Elsewhere Reward = 0
- Any action that takes you outside boundary, Reward = -1



# Gridworld – Q Learning - Properties

- Converges to optimal policy even if you are acting sub-optimally
- Also called “off-policy” learning
- Caveats:
  - Need to explore a lot
  - Need to make the learning rate small, but not decrease it too quickly
  - Exploration important or will stuck in local minimum





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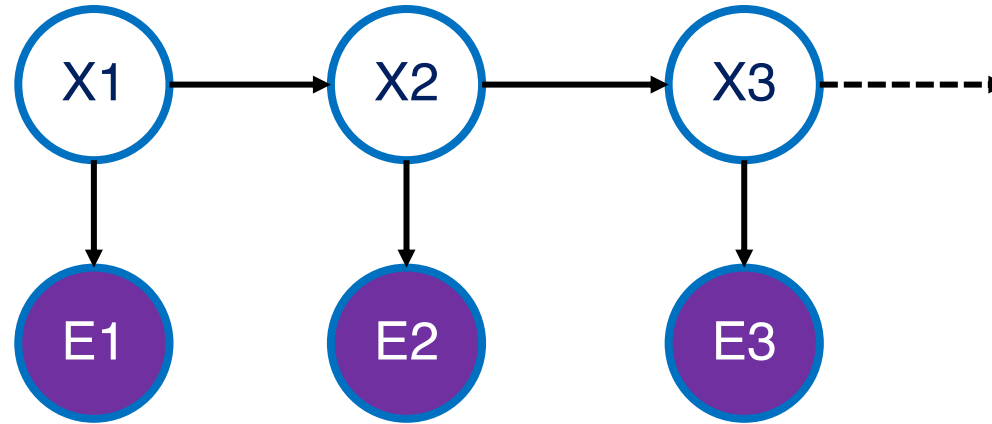
Email: [vigneshmdharan@iitj.ac.in](mailto:vigneshmdharan@iitj.ac.in)

# Hidden Markov Models

Probabilistic Reasoning through time

# Hidden Markov models: Intuition

Markov Model:



These states are always not observable in real-life scenarios

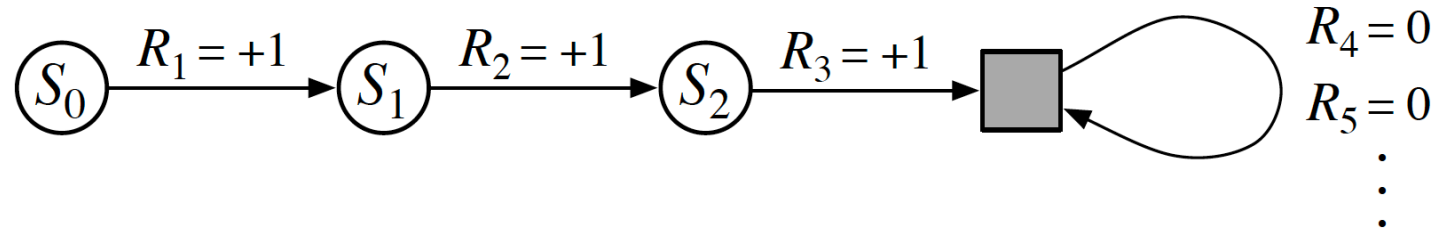
Hidden Markov Model:

Noisy evidence variables of the states

Observing the states indirectly, through measurement estimates

# Partially observable Markov Decision Processes (POMDPs)

Classic MDP – Assumption is environment is fully observable



When the environment is **partially observable** – the agent does not really know the state it is in.

Components of an MDP:  $P(s' | s, a)$ ,  $A(s)$  and  $R(s)$

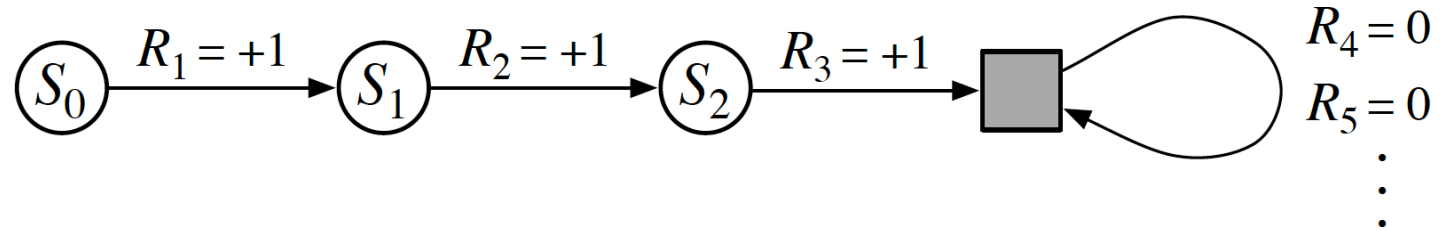
Additional components of a POMDP: Idea of a **sensor model** and a **belief state** ( $b(s)$ )

Sensor Model:  $P(e | s)$

Belief State:  $b(s') = \alpha P(e | s') \sum_{s'} P(s' | s, a) b(s')$

# Partially observable Markov Decision Processes (POMDPs)

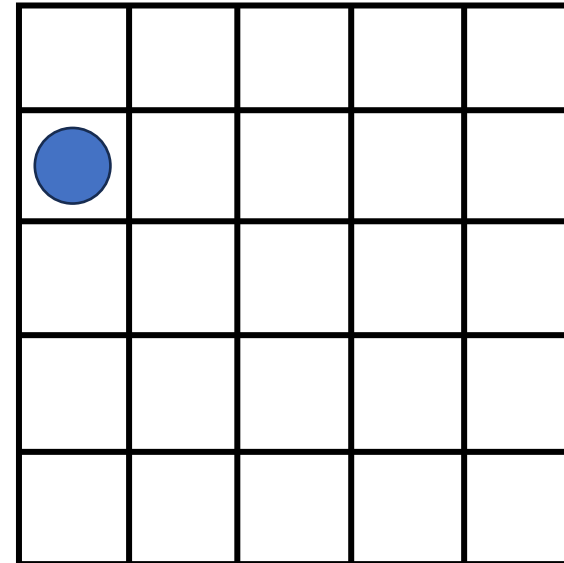
Classic MDP – Assumption is environment is fully observable



When the environment is **partially observable** – the agent does not really know the state it is in.

Sensor Model:  $P(e|s)$

Belief State:  $b(s') = \alpha P(e|s') \sum_{s'} P(s'|s, a) b(s')$





# Hidden Markov models: Find the agent

Let's look at an example scenario

# Markov vs Hidden Markov models

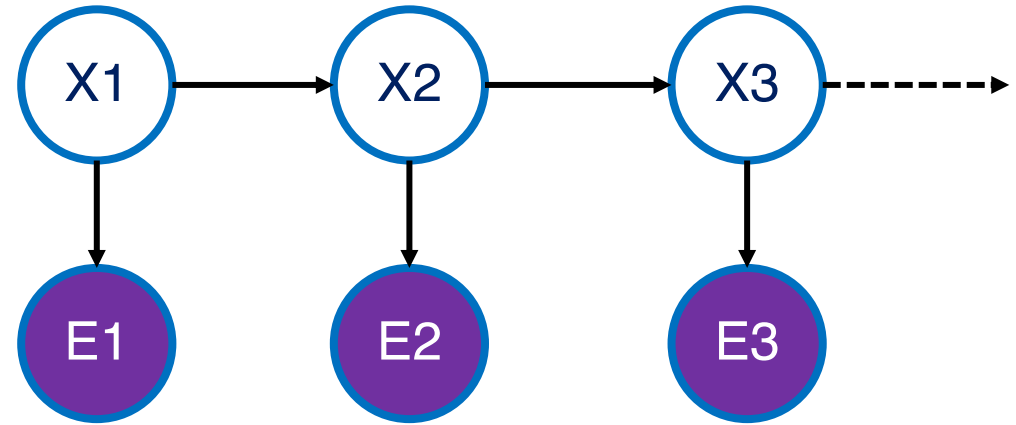
Markov Models:

Need observations to update your beliefs

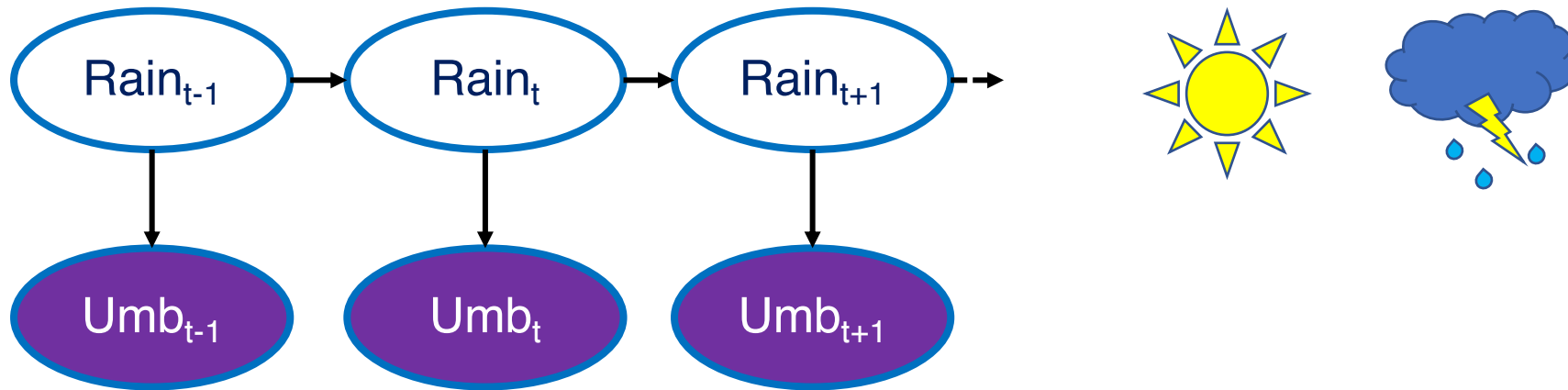
Hidden Markov Models:

Don't observe the states at all.

You observe the effects at each state.



# Classic HMM Example: Weather



Components of an HMM:

- Initial Probability distribution:  $P(X_1)$
- Transition:  $P(X_t | X_{t-1})$
- Emission:  $P(E_t | X_t)$

Transition probabilities

| $R_t$ | $R_{t+1}$ | $P(R_{t+1}   R_t)$ |
|-------|-----------|--------------------|
| +r    | +r        | 0.7                |
| +r    | -r        | 0.3                |
| -r    | +r        | 0.3                |
| -r    | -r        | 0.7                |

Emission probabilities

| $R_t$ | $U_t$ | $P(U_t   R_t)$ |
|-------|-------|----------------|
| +r    | +u    | 0.9            |
| +r    | -u    | 0.1            |
| -r    | +u    | 0.2            |
| -r    | -u    | 0.8            |

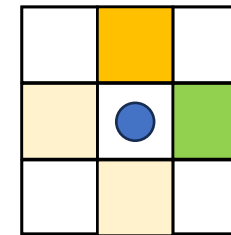
# HMM Example: Find the agent

Components of an HMM:

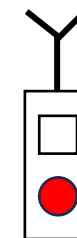
- Initial Probability distribution:  $P(X_1)$

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 0.01 | 0.00 | 0.04 | 0.00 | 0.06 | 0.00 | 0.03 | 0.00 | 0.01 | 0.00 |
| 0.00 | 0.03 | 0.00 | 0.05 | 0.00 | 0.05 | 0.00 | 0.03 | 0.00 | 0.02 |
| 0.03 | 0.00 | 0.04 | 0.00 | 0.05 | 0.00 | 0.04 | 0.00 | 0.03 | 0.00 |
| 0.00 | 0.04 | 0.00 | 0.04 | 0.00 | 0.04 | 0.00 | 0.03 | 0.00 | 0.02 |
| 0.02 | 0.00 | 0.03 | 0.00 | 0.05 | 0.00 | 0.04 | 0.00 | 0.02 | 0.00 |
| 0.00 | 0.01 | 0.00 | 0.03 | 0.00 | 0.06 | 0.00 | 0.04 | 0.00 | 0.01 |

- Transition:  $P(X_t|X_{t-1})$



- Emission:  $P(E_t|X_t)$



# Real-world HMM Examples

- **Speech Recognition:**
  - Observations: Acoustic Signals
  - States: Specific words being spoken
- **Machine Translation:**
  - Observations: Many words (ten to thousands)
  - States: Translation Options
- **Robot Tracking:**
  - Observations: Range readings/Sensors
  - States: Positions on a map/area

# Analogies to filtering/monitoring

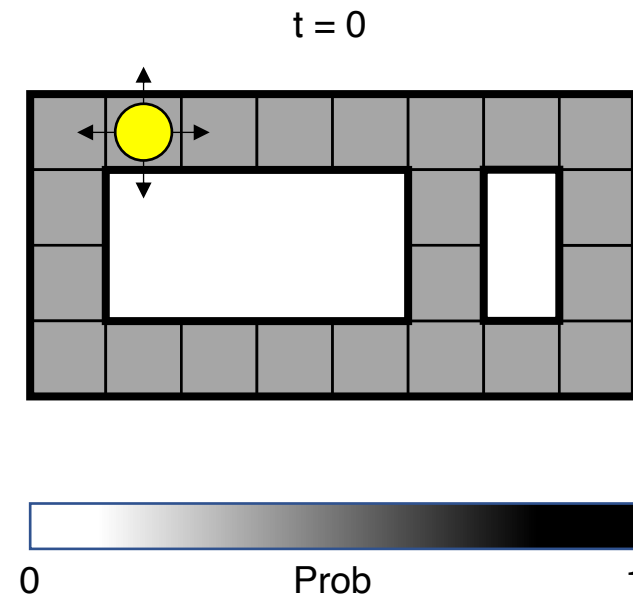
- Trying to track the distribution also called the belief state:

$$B_t(X) = P_t(X_t | e_1, e_2, \dots, e_t)$$

- We initially assume that the Belief state is uniform.
- As we get more and more information, we can start updating our belief state,  $B(X)$
- Kalman filters, formulated by Rudolf Kalman in the 60's – Space Program

# HMM Example: Robot Localization

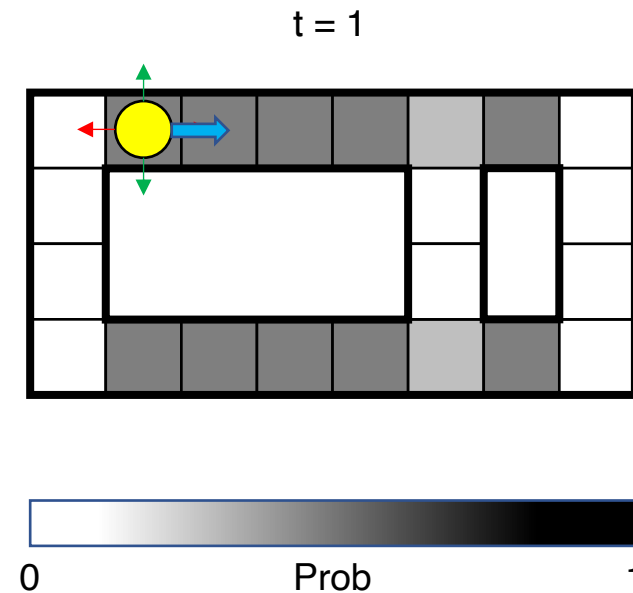
Example from  
Jacob Andreas  
Lecture, UCB



Let's consider two models:

- **Sensor Model:** able to detect whether there is a wall or not but not with complete certainty. Never more than 1 mistake.
- **Motion Model:** Robot moves to the right but may fail to move ahead with a small probability

# HMM Example: Robot Localization



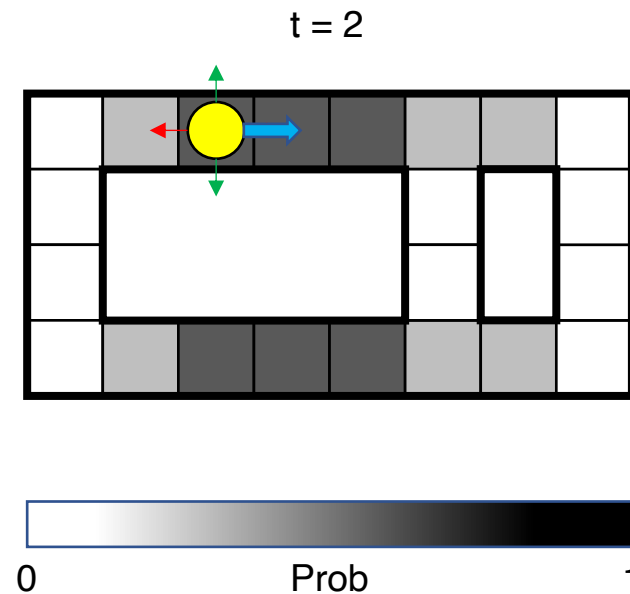
Q1: If the robot sits here, do you think it will become any more confident of its location?

Q2: If the robot sits here, do you think it will be totally confident about its position?

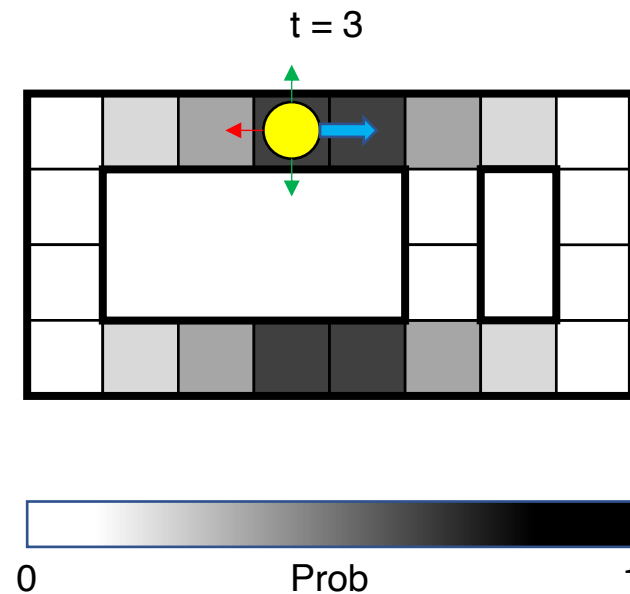
- Lighter grey: still some belief is there that the robot could be at that location, because the sensors might have made 1 mistake.



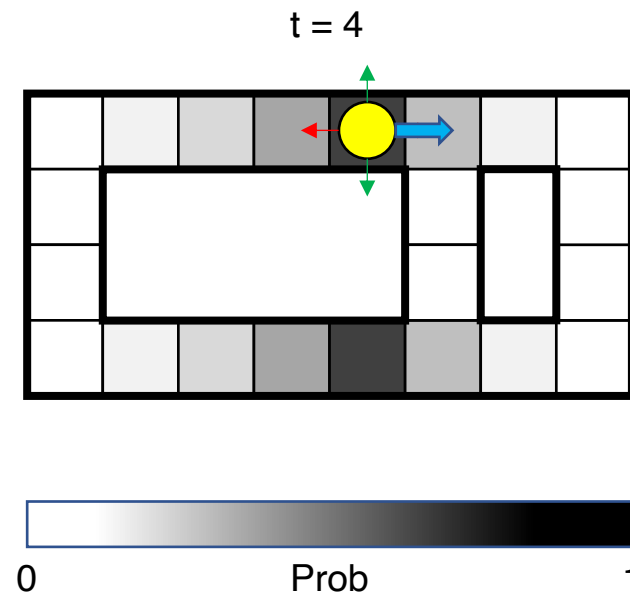
# HMM Example: Robot Localization



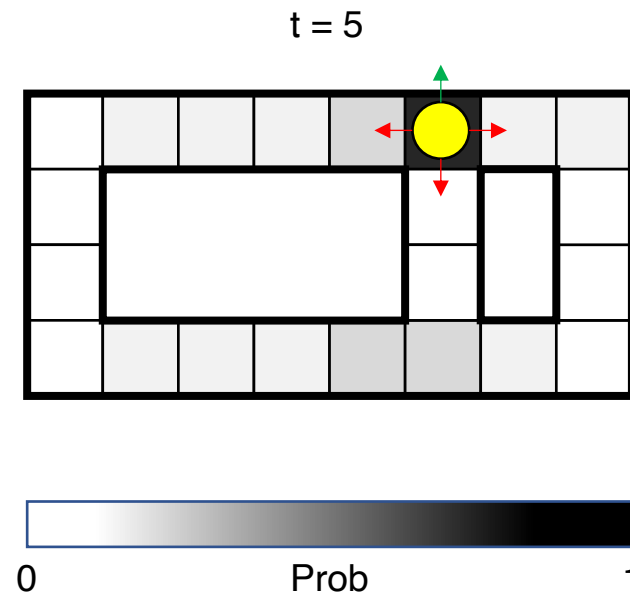
# HMM Example: Robot Localization



# HMM Example: Robot Localization



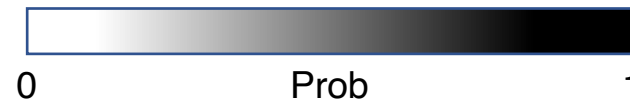
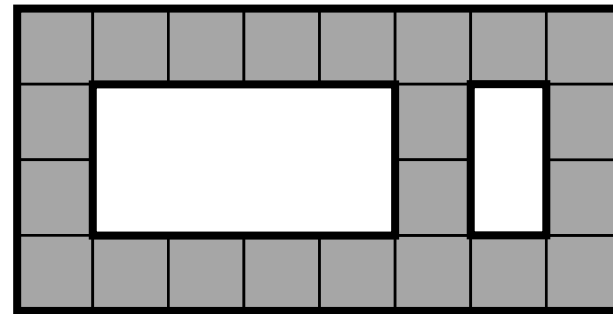
# HMM Example: Robot Localization



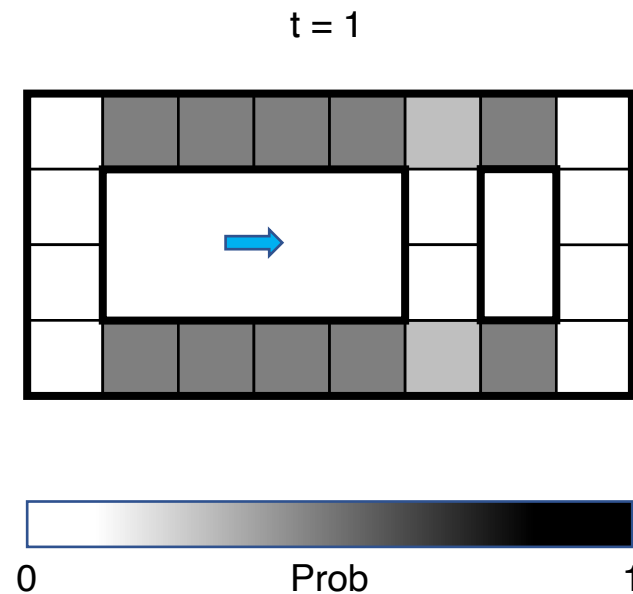
# HMM Example: Robot Localization

What the robot actually sees

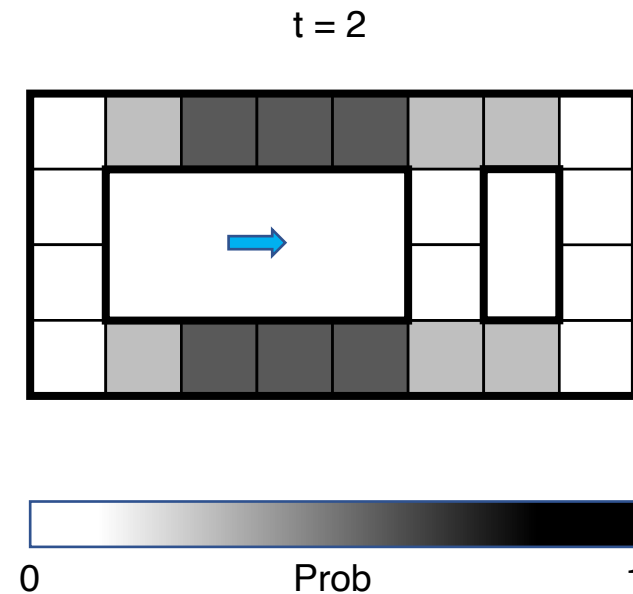
$t = 0$



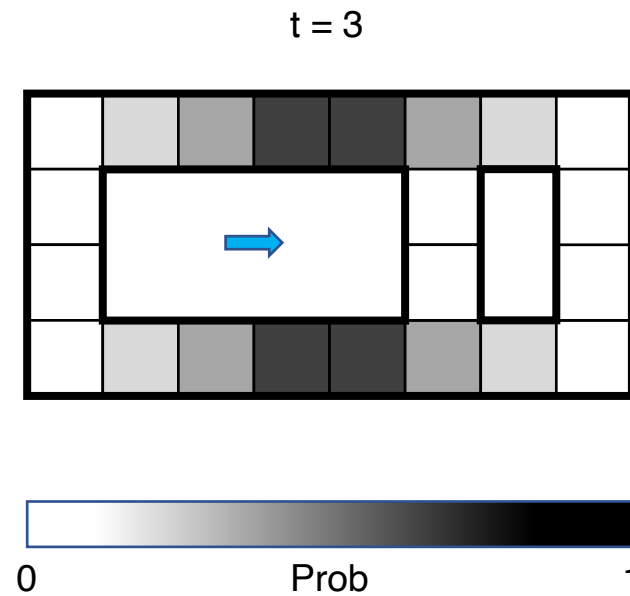
# HMM Example: Robot Localization



# HMM Example: Robot Localization

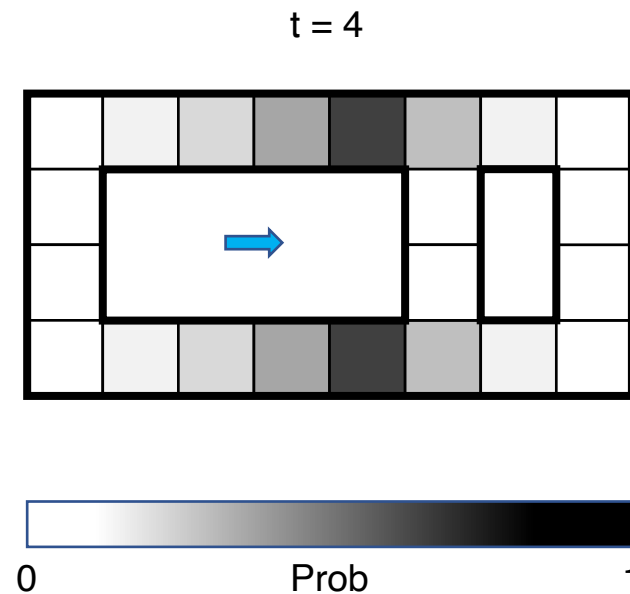


# HMM Example: Robot Localization

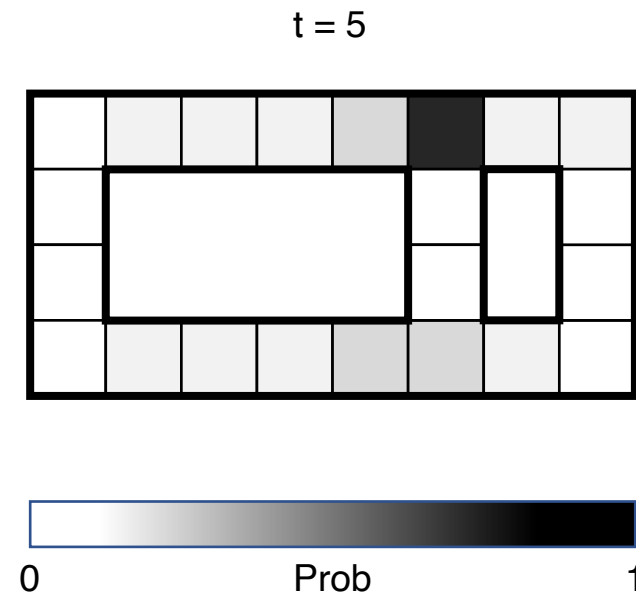




# HMM Example: Robot Localization



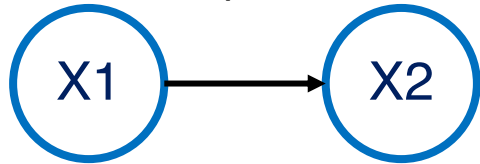
# HMM Example: Robot Localization



# HMM Inference: Base Cases

Two things to solve  $B(X_t) = P(X_t|e_{1:t})$

What happens to that belief as  
time passes

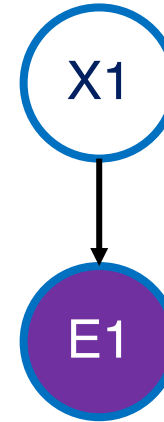


$P(X_2)$

$$P(X_2) = \sum_{X_1} P(X_1, X_2)$$

$$P(X_2) = \sum_{X_1} P(X_1)P(X_2|X_1)$$

How does our belief get updated  
as we get new evidence



$$P(X_1|e_1) = \frac{P(e_1|X_1)P(X_1)}{P(e_1)} \\ \propto P(e_1|X_1)P(X_1)$$

# HMM Inference: Passage of Time

Belief Distribution  $B(X_t) = P(X_t|e_{1:t})$

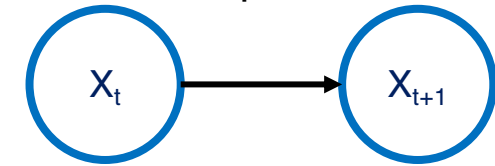
After one time-step

$$\begin{aligned} B'(X_{t+1}) &= P(X_{t+1}|e_{1:t}) = \sum_{X_t} P(X_{t+1}, X_t|e_{1:t}) \\ &= \sum_{X_t} P(X_{t+1}|X_t, e_{1:t}) P(X_t|e_{1:t}) \\ &= \sum_{X_t} P(X_{t+1}|X_t) B(X_t) \end{aligned}$$

So, we can write

$$B'(X_{t+1}) = \sum_{X_t} P(X_{t+1}|X_t) B(X_t)$$

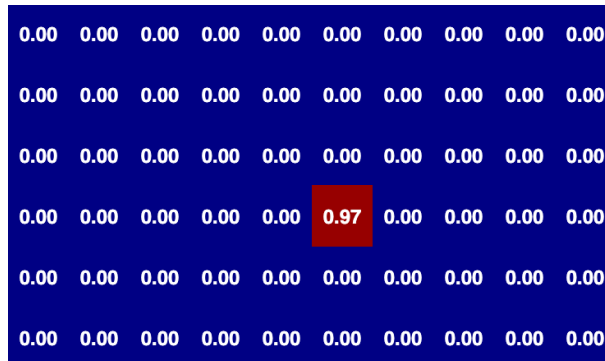
What happens to that belief as time passes



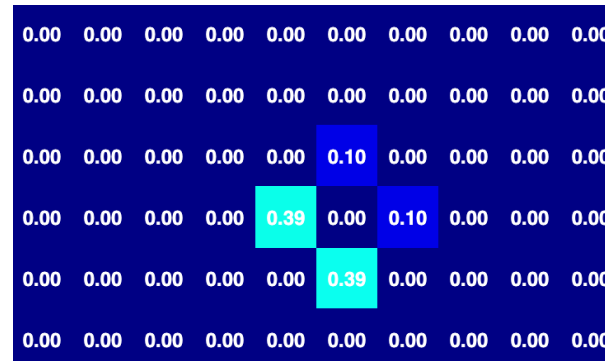
# HMM Inference: Passage of Time

Find the Agent

$t = 1$

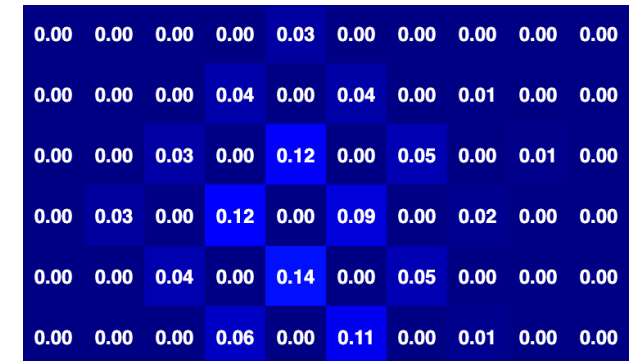


$t = 2$



.....

$t = 5$



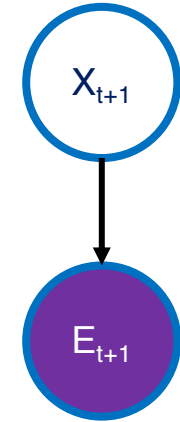
# HMM Inference: Belief update after new observation

Belief Distribution  $B'(X_{t+1}) = P(X_{t+1}|e_{1:t})$

After the observation

$$\begin{aligned} B(X_{t+1}) &= P_t(X_{t+1}|e_{1:t+1}) \propto P(e_{t+1}|X_{t+1}, e_{1:t})P(X_{t+1}|e_{1:t}) \\ &\propto P(e_{t+1}|X_{t+1})P(X_{t+1}|e_{1:t}) \\ &\propto P(e_{t+1}|X_{t+1})B'(X_{t+1}) \end{aligned}$$

How does our belief get updated as we get new evidence



So, beliefs get reweighted by the likelihood of evidence

# HMM Inference: Belief after observations

Before Observation

|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 |
| 0.00 | 0.00 | 0.00 | 0.35 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.01 |
| 0.00 | 0.00 | 0.35 | 0.00 | 0.10 | 0.00 | 0.01 | 0.00 | 0.01 | 0.00 |
| 0.00 | 0.00 | 0.00 | 0.09 | 0.00 | 0.03 | 0.00 | 0.01 | 0.00 | 0.00 |

After Observation

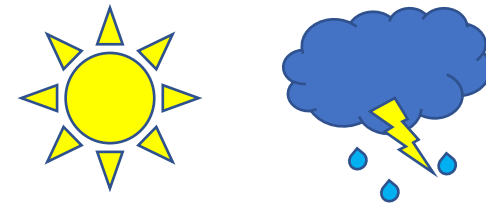
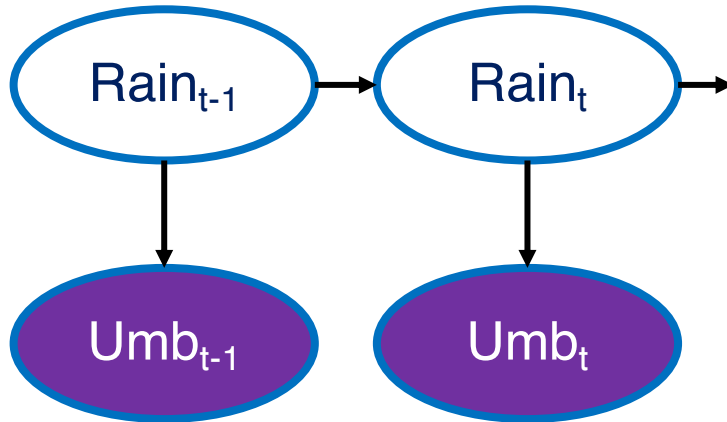
|      |      |      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|------|------|
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 |
| 0.00 | 0.00 | 0.00 | 0.84 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 |
| 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.01 | 0.00 | 0.01 | 0.00 |
| 0.00 | 0.00 | 0.00 | 0.07 | 0.00 | 0.02 | 0.00 | 0.01 | 0.00 | 0.00 |

$$B(X_{t+1}) \propto P(e_{t+1}|X_{t+1})B'(X_{t+1})$$

# Classic HMM Example: Weather Inference

$$\begin{array}{cc} \begin{array}{l} B'(+r) = 0.5 \\ B'(-r) = 0.5 \end{array} & \begin{array}{l} B'(+r) = 0.6 \\ B'(-r) = 0.4 \end{array} \\ \downarrow & \downarrow \\ \begin{array}{l} B(+r) = 0.8 \\ B(-r) = 0.2 \end{array} & \begin{array}{l} B(+r) = 0.9 \\ B(-r) = 0.1 \end{array} \end{array}$$

Diagram illustrating the relationship between two sets of probabilities,  $B'$  and  $B$ , for weather states (+r and -r). Arrows indicate transitions or dependencies between the two sets.







# Advanced Artificial Intelligence

Instructors: Dr. Vignesh Muralidharan and Dr. Dipanjan Roy  
School of AI and Data Science  
IIT Jodhpur

Email: [vigneshmdharan@iitj.ac.in](mailto:vigneshmdharan@iitj.ac.in)

# Time Series Analysis

Kalman Filters

# Time-series data

Navigation control

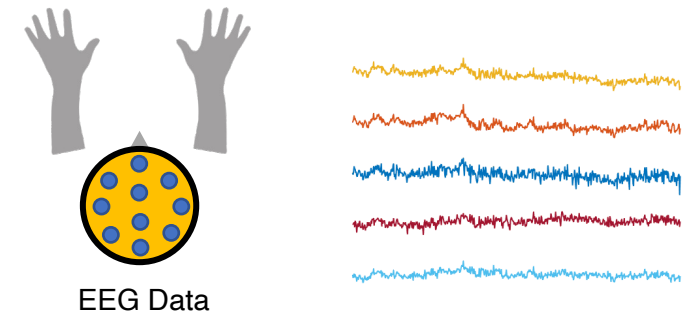


aircraftnerds.com

Automation control



Brain-Machine control

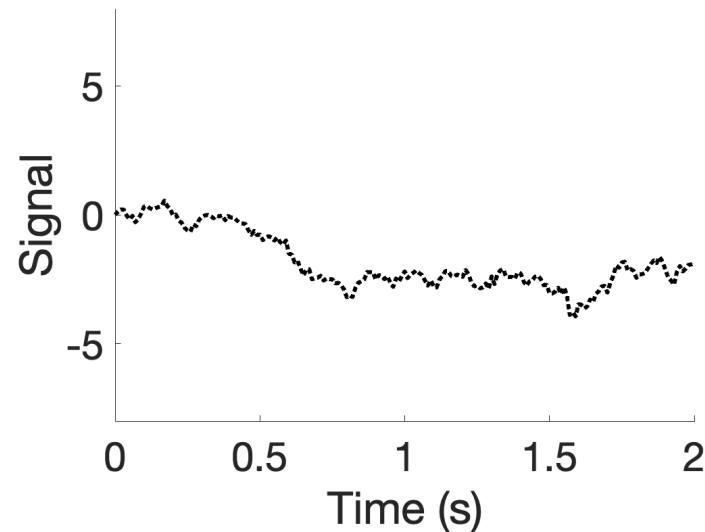


# Time-series analysis

Suppose we have a time-series signal as follows:

$$a_{t+1} = a_t + \eta_t \quad \eta_t \sim N(0, \sigma_\eta^2)$$

Random Walk



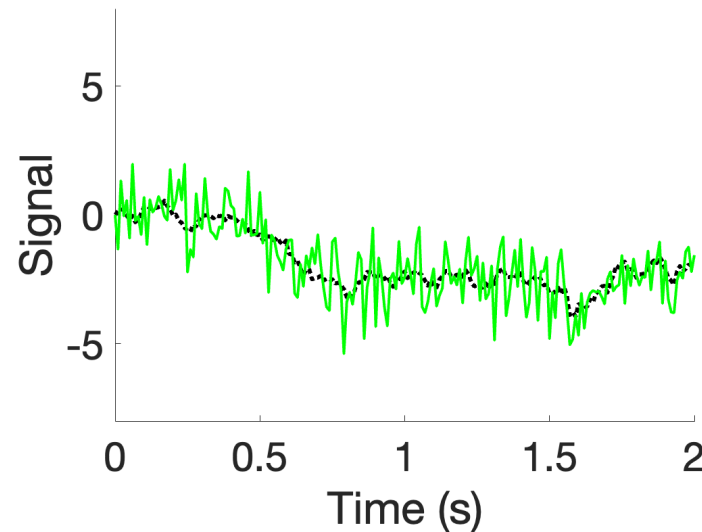
# Time-series analysis

Introducing a second time-series as follows:

$$y_t = a_t + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

Random Walk

If we can observe only  $y$ ,  
what can we say about  
the  $a$ ?



# Time-series analysis: State Space Models

State Equation:

$$a_{t+1} = a_t + \eta_t \quad \eta_t \sim N(0, \sigma_\eta^2)$$

Observation (Measurement) Equation:

$$y_t = a_t + \varepsilon_t \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

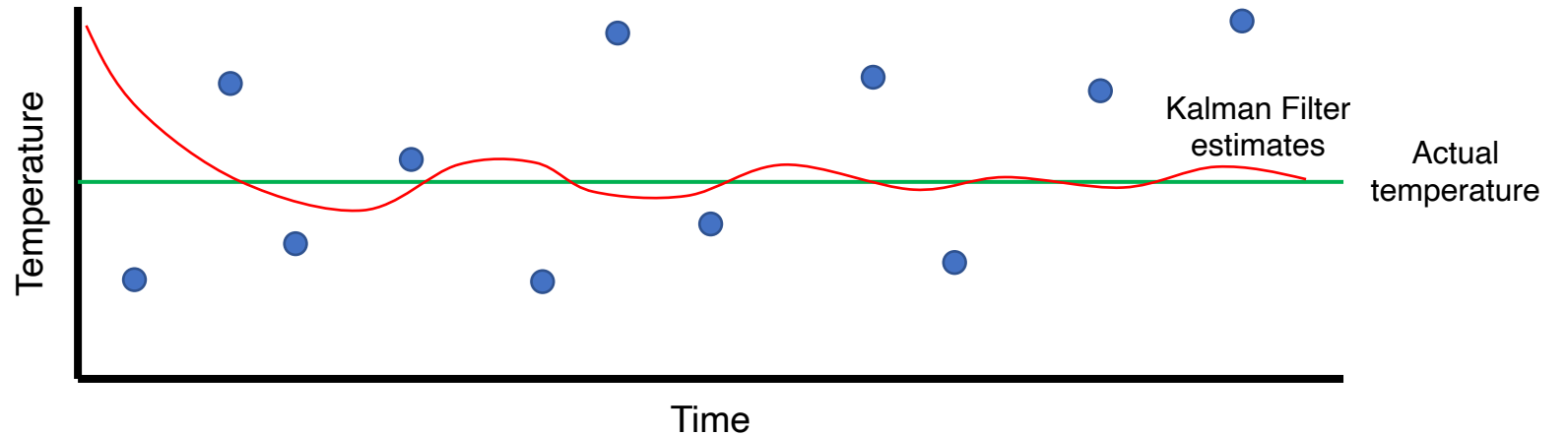
Some examples:

| State              | Observation                                     |
|--------------------|-------------------------------------------------|
| Signal             | Signal + Noise                                  |
| Position, Velocity | Measurement from sensors (Gyro, Accelerometers) |
| Commodity Prices   | Future Prices                                   |
| Cursor Position    | Finger position on a trackpad                   |

# Time-series analysis: Kalman Filters

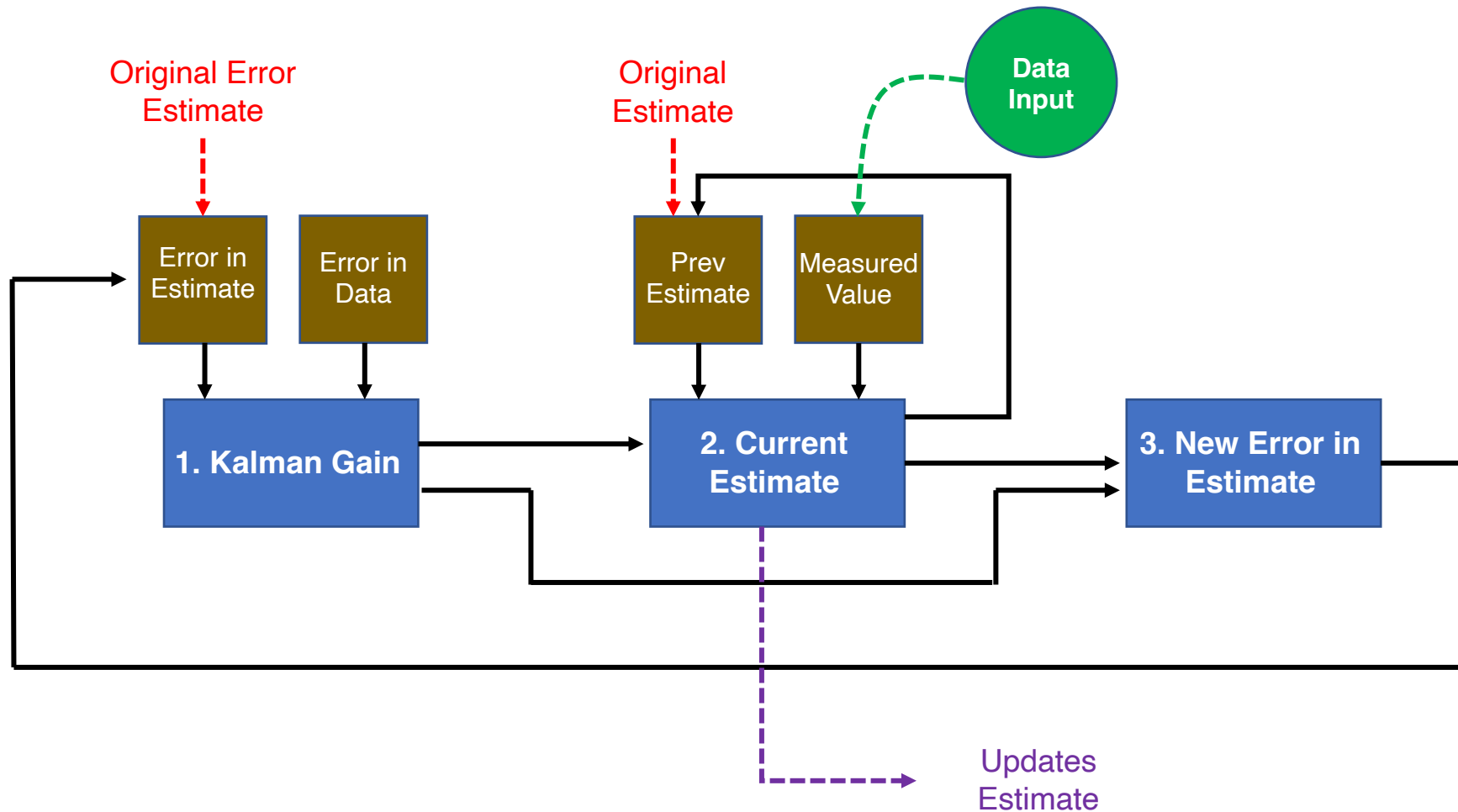
Iterative or Recursive Process: Estimate the true value of the state of the system from noisy/uncertain measurements

Sensing temperature



Can be extended to complicated setups: Tracking an airplane for e.g.

# Kalman Filter: Flow diagram

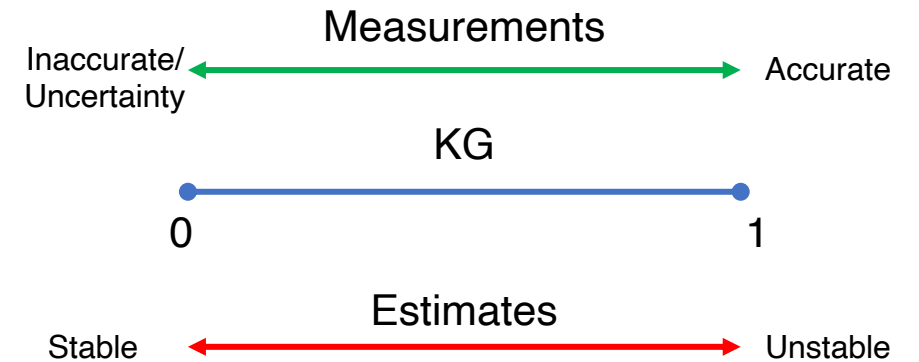
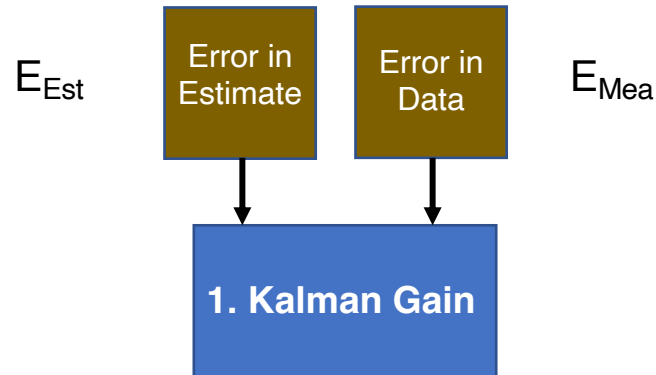




# Kalman Filter: The Kalman Gain

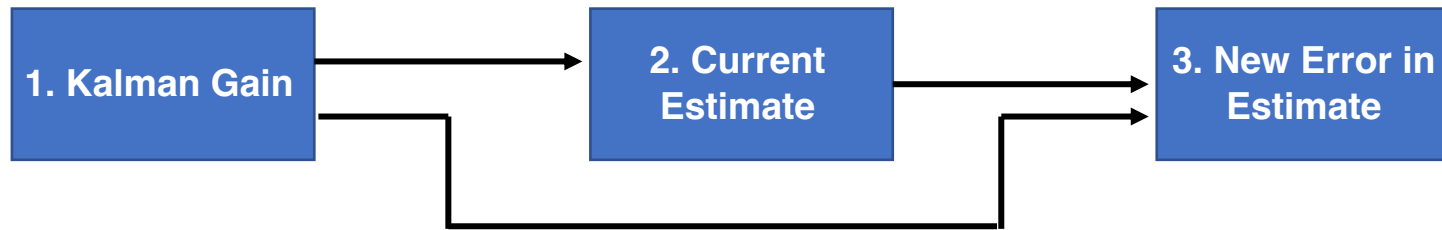
$$KG = \frac{E_{Est}}{E_{Est} + E_{Mea}}$$

$$0 \leq KG \leq 1$$



Update in current estimate:  $Est_t = Est_{t-1} + KG(Mea - Est_{t-1})$

# Kalman Filter: Our 3 main components



$$KG = \frac{E_{Est}}{E_{Est} + E_{Mea}}$$

$$Est_t = Est_{t-1} + KG(Mea - Est_{t-1})$$

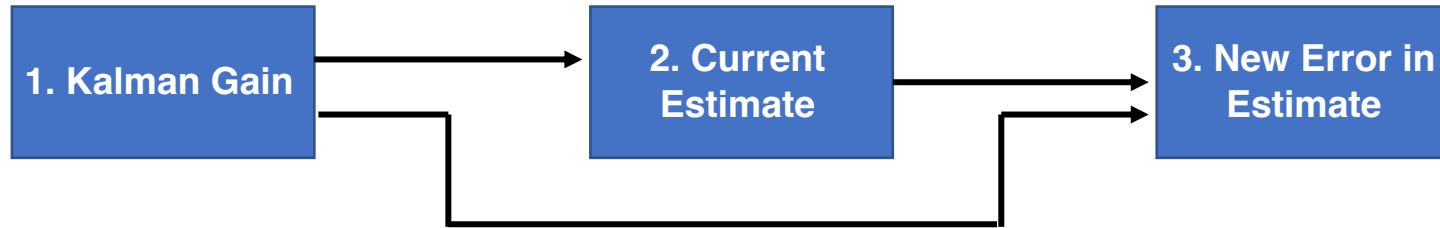
$$E_{Est\_t} = \frac{(E_{Mea})(E_{Est\ t-1})}{E_{Mea} + E_{Est\ t-1}}$$



$$E_{Est\_t} = (1 - KG)E_{Est\_t-1}$$

# Kalman Filter: Numerical Example

Further  
Measurements =  
71, 70, 74



True Temperature = 72

Initial Estimate = 68

Initial Error<sub>Est</sub> = 2

Initial Measurement = 75

Initial Error<sub>Mea</sub> = 4

$$KG = \frac{E_{Est}}{E_{Est} + E_{Mea}}$$

$$Est_t = Est_{t-1} + KG(Mea - Est_{t-1})$$

$$E_{Est\_t} = (1 - KG)E_{Est\_t-1}$$